# Brazilian Strategy for Managing the Risk of Foreign Exchange Rate Exposure during a Crisis

**Prof. Dr. Antonio Francisco de A. da Silva Jr.**

# Summary

- Inflation target
- Foreign exchange rate regimes
- International reserves
- The Brazilian strategies to deal with financial crisis

# Summary

- **Inflation target**
- Foreign exchange rate regimes
- International reserves
- The Brazilian strategies to deal with financial crisis

# Does inflation matter?

- High and volatile inflation rates generate distortions that lead to increased risks and negatively affect investment. These distortions **shorten the planning horizons** of households, companies and governments and erode business confidence.

- High inflation rates **subtract the purchasing power of wages** and transfers, with negative repercussions on household's confidence and consumption.

- High inflation rates **produce inefficient price dispersion and reduce the informational value from prices** that contributes to the efficient allocation of resources.

- High and volatile inflation also has **regressive distributive effects**. The less favored groups of the population, which generally have more restricted access to instruments to protect them from the loss of the currency's purchasing power, benefit the most from price stability.

Central Bank of Brazil (2022). Inflation report, vol 24, n. 2.

# Inflation Target

The inflation target framework is an alternative to others monetary policy approaches like those based on an exchange rate peg or some monetary aggregate as an intermediate target.

The inflation target framework was first adopted in the early 1990s by New Zealand, Canada, the United Kingdom and Sweden.

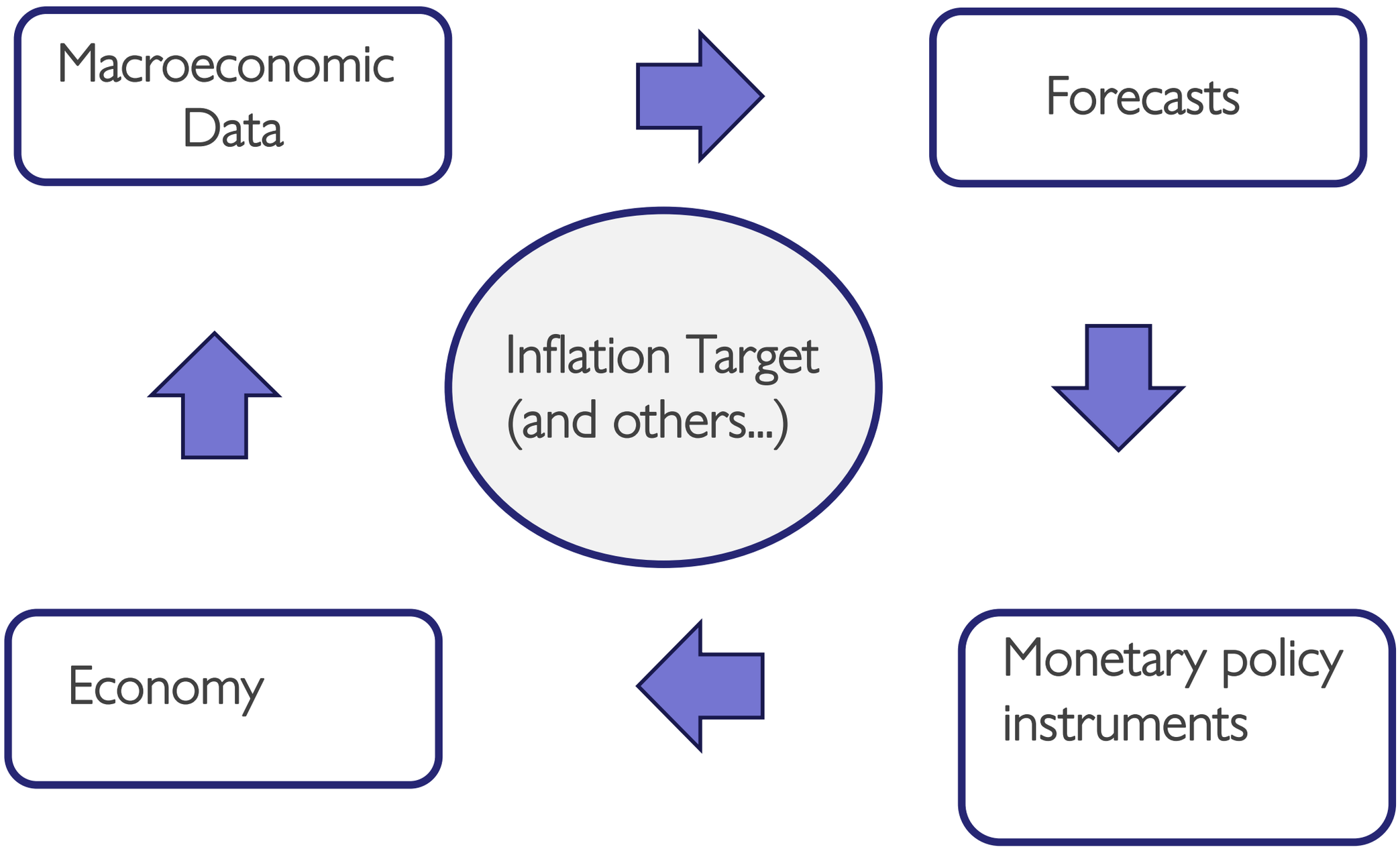
The framework stablishes an explicit target for inflation. In some approaches other goals are pursued like full employment and exchange rate volatility, but there are no explicit targets for these goals.

# Other monetary arrangements: Exchange rate peg and monetary aggregates

Exchange rate peg relies on the exchange rate impact in the inflation. Therefore, the central bank search an exchange rate level to avoid high inflation.

Monetary aggregates approaches are based on the amount of money in the economy. The central bank controls the monetary aggregate in order to keep inflation under control.

# Inflation Target



# Inflation target – Taylor Rule

A central bank tights monetary policy when it raises the nominal interest rate by more than an increase in inflation. On the other hand, the central bank may adopt an easy monetary policy lowering interest rates.

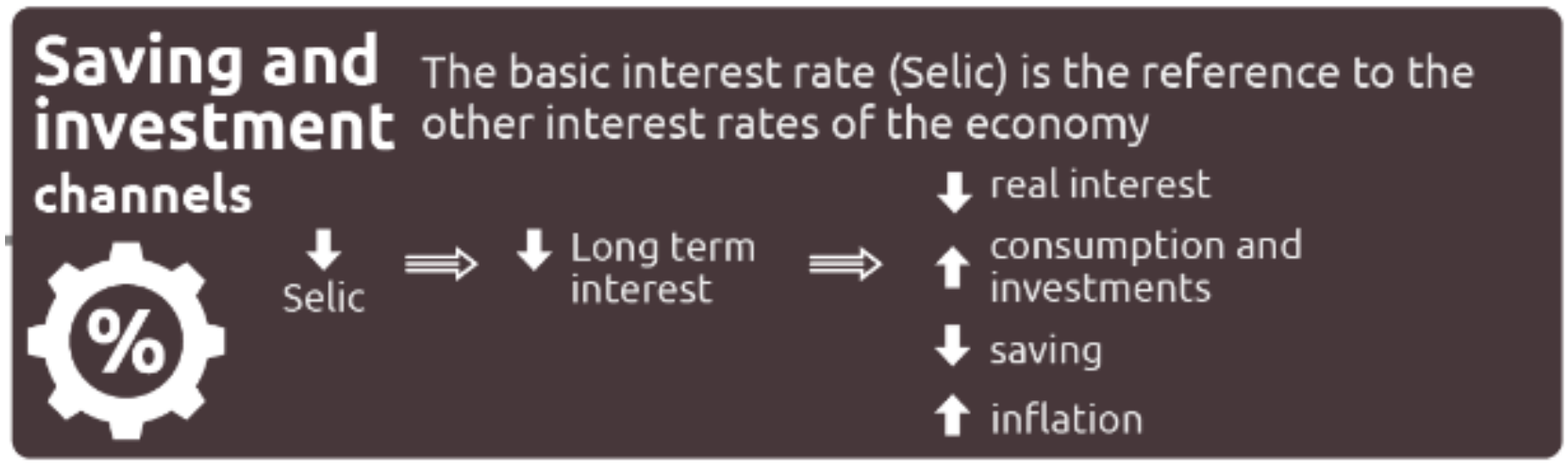
# Questions

1. How may interest rates affect inflation?

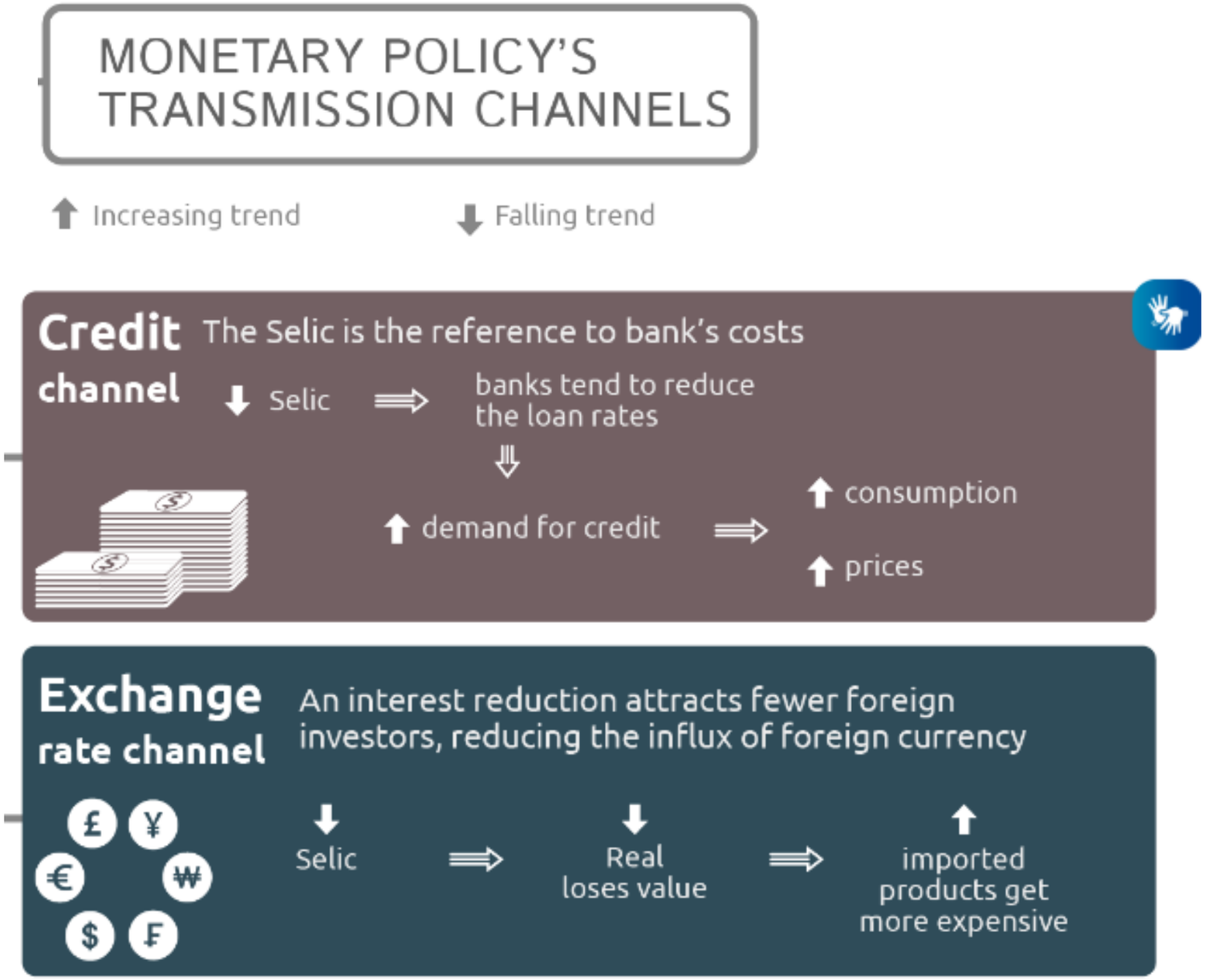
2. Does the mechanism differ between countries?

# MONETARY POLICY'S TRANSMISSION CHANNELS

⬆ Increasing trend ⬇ Falling trend



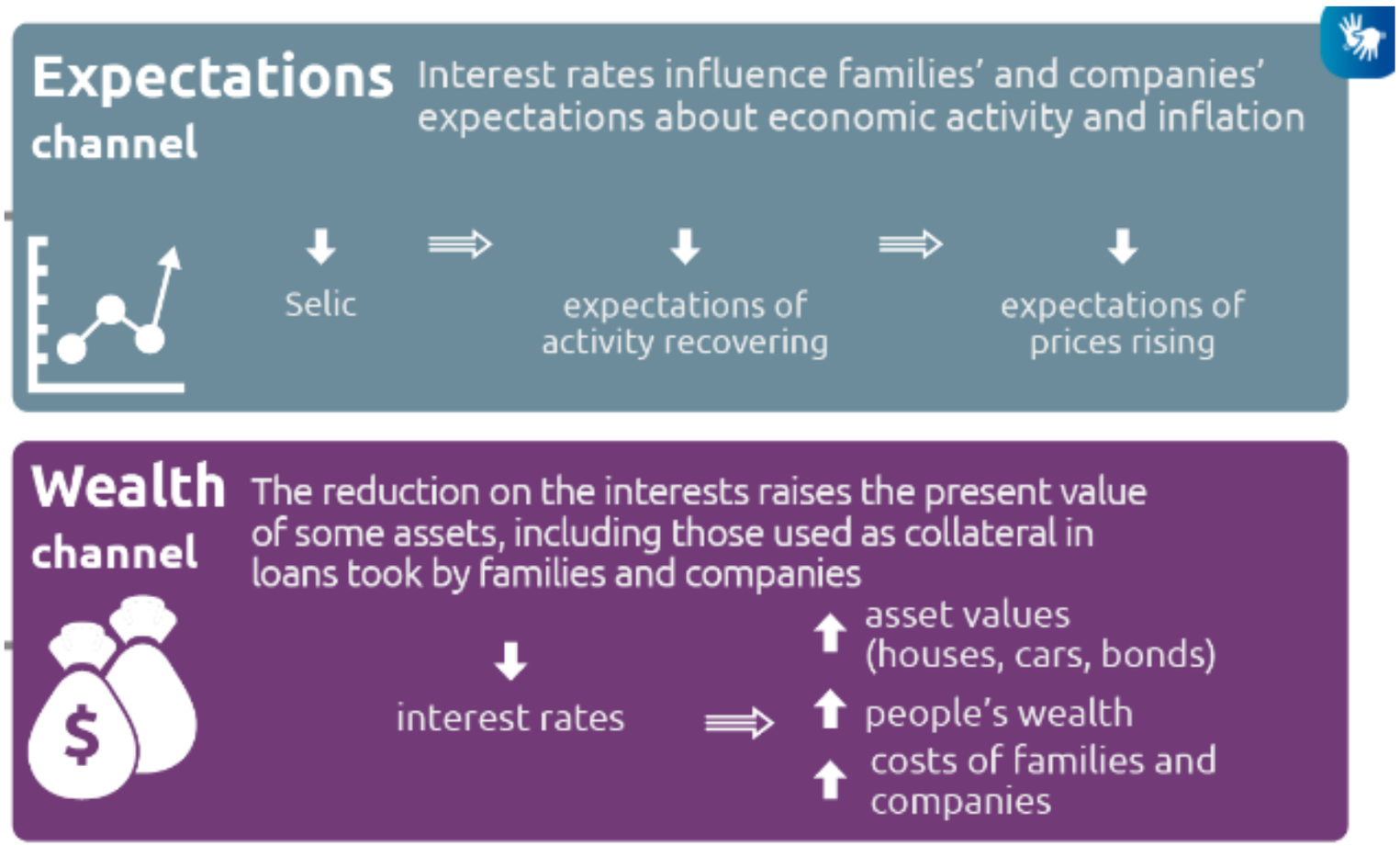

Source: https://www.bcb.gov.br/en/monetarypolicy/transmission_channels




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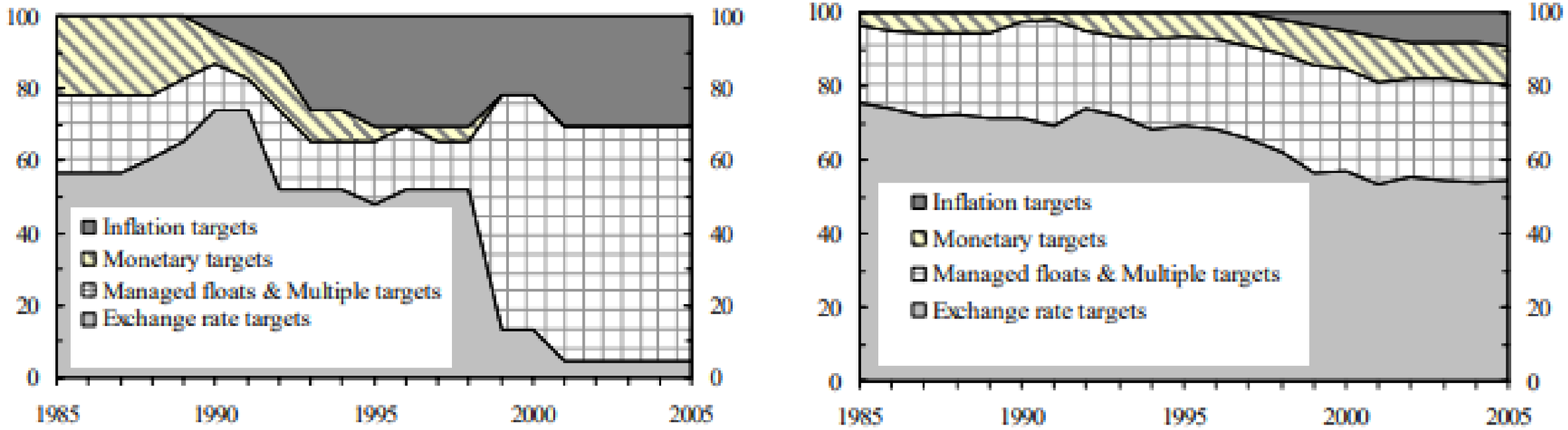
# MONETARY POLICY'S TRANSMISSION CHANNELS

⬆ Increasing trend ⬇ Falling trend




Source: https://www.bcb.gov.br/en/monetarypolicy/transmission_channels

Figure 1. Evolution of Monetary Policy Regimes, 1985-2005




Source: IMF (2006). Inflation Targeting and the IMF

Table 1. Inflation Targeters 1/

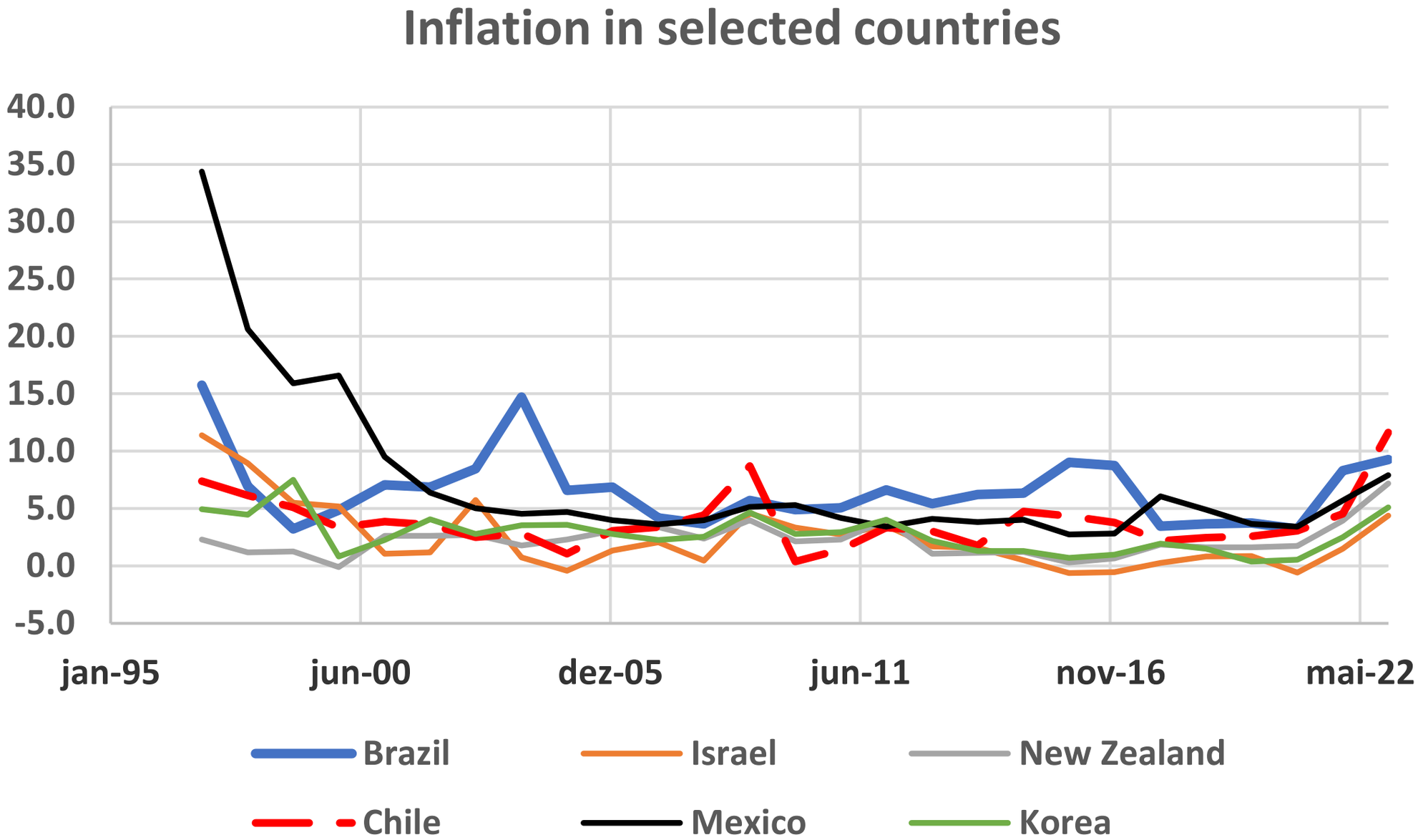
| | Inflation Targeting Adoption Date | Inflation Rate at Start (percent) | Unique Numeric Target = Inflation | Current Inflation Target (percent) | Forecast Process | Publish Forecast |
|---|---|---|---|---|---|---|
| Emerging market countries | | | | | | |
| Israel | 1997Q2 | 8.5 | Y | 1–3 | Y | Y |
| Czech Rep. | 1998Q1 | 13.1 | Y | 3 (+/- 1) | Y | Y |
| Poland | 1998Q4 | 9.9 | Y | 2.5 (+/- 1) | Y | Y |
| Brazil | 1999Q2 | 3.3 | Y | 4.5 (+/- 2.0) | Y | Y |
| Chile | 1999Q3 | 2.9 | Y | 2–4 | Y | Y |
| Colombia | 1999Q3 | 9.3 | Y | 5 (+/- 0.5) | Y | Y |
| South Africa | 2000Q1 | 2.3 | Y | 3–6 | Y | Y |
| Thailand | 2000Q2 | 1.7 | Y | 0–3.5 | Y | Y |
| Korea | 2001Q1 | 3.2 | Y | 2.5–3.5 | Y | Y |
| Mexico | 2001Q1 | 8.1 | Y | 3 (+/-1) | Y | N |
| Hungary | 2001Q2 | 10.5 | Y | 3.5 (+/- 1) | Y | Y |
| Peru | 2002Q1 | -0.8 | Y | 2.5 (+/- 1) | Y | Y |
| Philippines | 2002Q1 | 3.8 | Y | 5–6 | Y | Y |
| Slovak Rep. | 2005Q1 | 3.2 | Y | 3.5 (+/- 1) | Y | Y |
| Indonesia | 2005Q3 | 7.8 | Y | 5.5 (+/- 1) | Y | Y |
| Romania | 2005Q3 | 8.8 | Y | 7.5 (+/- 1) | Y | Y |
| Industrial countries | | | | | | |
| New Zealand | 1990Q1 | 7.0 | Y | 1–3 | Y | Y |
| Canada | 1991Q1 | 6.2 | Y | 1–3 | Y | Y |
| United Kingdom | 1992Q4 | 3.6 | Y | 2 | Y | Y |
| Sweden | 1993Q1 | 4.8 | Y | 2 (+/- 1) | Y | Y |
| Australia | 1993Q2 | 1.9 | Y | 2–3 | Y | Y |
| Iceland | 2001Q1 | 3.9 | Y | 2.5 | Y | Y |
| Norway | 2001Q1 | 3.7 | Y | 2.5 | Y | Y |

Source: IMF (2006). Inflation Targeting and the IMF

# Inflation in selected countries

| Year | Brazil | Israel | New Zealand | Chile | Mexico | Korea |
|---|---|---|---|---|---|---|
| 1980 | | 131,0 | 17,2 | 35,1 | 26,4 | 28,7 |
| 1981 | 101,7 | 116,8 | 15,4 | 19,7 | 27,9 | 21,4 |
| 1982 | 100,5 | 120,4 | 16,2 | 9,9 | 58,9 | 7,2 |
| 1983 | 135,0 | 146,0 | 7,3 | 27,3 | 101,9 | 3,4 |
| 1984 | 192,1 | 373,2 | 6,2 | 19,9 | 65,4 | 2,3 |
| 1985 | 226,0 | 308,8 | 15,4 | 30,7 | 57,7 | 2,5 |
| 1986 | 147,1 | 48,0 | 13,2 | 19,5 | 86,2 | 2,7 |
| 1987 | 228,3 | 19,1 | 15,7 | 19,9 | 131,8 | 3,0 |
| 1988 | 629,1 | 16,4 | 6,4 | 14,7 | 114,2 | 7,1 |
| 1989 | 1430,7 | 20,3 | 5,7 | 17,0 | 20,0 | 5,7 |
| 1990 | 2947,7 | 17,3 | 6,1 | 26,0 | 26,7 | 8,6 |
| 1991 | 432,8 | 19,0 | 2,6 | 21,8 | 22,7 | 9,3 |
| 1992 | 952,0 | 12,0 | 1,0 | 15,4 | 15,5 | 6,2 |
| 1993 | 1927,4 | 11,0 | 1,3 | 12,7 | 9,8 | 4,8 |
| 1994 | 2075,9 | 12,3 | 1,7 | 11,4 | 7,0 | 6,3 |
| 1995 | 66,0 | 9,9 | 3,8 | 8,2 | 35,0 | 4,5 |

Source: Author with FRED data




Source: Author with FRED data
Updated: Jun/2023

# Inflation target and Non-inflation target countries

| IT countries (15) | Inflation targeting adoption year | 2007–2009 Average inflation rate (%) | Non-IT countries (39) | 2007–2009 Average inflation rate (%) | | 2007–2009 Average inflation rate (%) |
|---|---|---|---|---|---|---|
| Czech Republic | 1997 | 6.4 | Algeria | 4.9 | Kazakhstan | 17.1 |
| Poland | 1998 | 4.2 | Angola | 12.5 | Kenya | 26.2 |
| Brazil | 1999 | 5.7 | Argentina | 8.6 | Lao PDR | 7.6 |
| Chile | 1999 | 8.7 | Bangladesh | 8.9 | Malaysia | 5.4 |
| Colombia | 1999 | 7.0 | Belarus | 14.8 | Morocco | 3.7 |
| South Africa | 2000 | 10.1 | Benin | 7.9 | Nigeria | 11.6 |
| Thailand | 2000 | 5.5 | Bolivia | 14.0 | Pakistan | 20.3 |
| South Korea | 2001 | 4.7 | Bulgaria | 12.3 | Russia | 14.1 |
| Mexico | 2001 | 5.1 | Cambodia | 25.0 | Senegal | 7.3 |
| Hungary | 2001 | 6.0 | Cameroon | 5.3 | Sri Lanka | 22.6 |
| Peru | 2002 | 5.8 | China | 5.9 | Sudan | 14.3 |
| Philippines | 2002 | 8.3 | Côte d'Ivoire | 6.3 | Syria | 15.7 |
| Indonesia | 2005 | 10.2 | Ecuador | 8.4 | Tanzania | 10.3 |
| Guatemala | 2005 | 11.4 | Egypt | 18.3 | Ukraine | 25.2 |
| Romania | 2005 | 7.9 | El Salvador | 6.7 | Venezuela | 27.1 |
| | | | Haiti | 15.3 | Vietnam | 23.1 |
| | | | Honduras | 11.4 | Yemen | 19.0 |
| | | | India | 8.3 | Zambia | 12.4 |
| | | | Iran | 25.4 | Zimbabwe | 157 |
| | | | | | Tunisia | 4.3 |

**Note(s):** Country selection is based on Lin and Ye (2009)'s approach
**Source(s):** IMF (2019), The World Bank Data, Central banks' data

**Table 1.**
Sample countries

Source: Duong, T. H. (2021). Inflation targeting and economic performance over the crisis: evidence from emerging market economies. Asian Journal of Economics and Banking

# Inflation target and economic performance

Table 3. Gains/Losses from Different Regimes

| Variables | Inflation Targeting | Fixed Exchange Rate (currency board/ dollarization/ peg) | Monetary Targeting | Other Regime |
|---|---|---|---|---|
| CPI inflation | -4.8** | -0.1 | 1.8 | 4.5 |
| Volatility of CPI inflation | -3.6** | 1.1 | 2.9 | 0.8 |
| Volatility of real output growth | -0.6 | 0.0 | 1.0 | -0.1 |
| Volatility of output gap | -0.01** | -0.001 | 0.01 | 0.003 |
| 5-year inflation forecast | -2.7** | 3.1 | n.a. | -0.1 |
| Volatility of 5-year inflation forecast | -2.1** | 2.3 | n.a. | -0.0 |
| 6–10-year inflation forecast | -2.2** | 2.2 | n.a. | -0.1 |
| Volatility of 6–10-year inflation forecast | -1.7*** | 1.2 | n.a. | -0.0 |

Sources: IMF, *International Financial Statistics;* and IMF staff calculations.
Note: Gains/losses measured in percentage points changes in variables. One, two, and three asterisks denote statistical significance at the 10, 5, and 1 percent level, respectively. Volatility is measured by the standard deviation of the variable.

Source: IMF (2006). Inflation Targeting and the IMF
Source: IMF (2006). Inflation Targeting and the IMF

# Technical requirements for inflation targeting

- **Institutional independence**: central bank must have autonomy

- **Analytical capabilities and infrastructure**: data requirements are more demanding

- **Economic structure**: prices deregulation and minimal dollarization

- **Healthy financial system**: banking system should be sound, and capital markets well-developed

# Questions

1. Should central banks pursue more than one objective in conducting monetary policy?

2. Which are monetary policy operations in an inflation target program?

# Inflation target – Central Bank mission (Brazil)

## Strategic objectives 2020-2023 (www.bcb.gov.br)

- **To meet the inflation targets set by the National Monetary Council (CMN) and — without harm to the price stability — to smooth fluctuations in economic activity, and to promote full employment.**
- Maintain the national financial stability
- Facilitate access to financial markets
- Improve the transparency, quality and flow of information of both the financial market and Banco Central do Brasil
- To foster competition within the Financial and Payment Systems
- Stimulate savings and a conscious market participation
- Promote sustainable finances and contribute to the reduction of socio-environmental and climatic risks within the economy and the Financial System
- To enhance the Banco Central do Brasil's capacity by means of innovations, technology, efficient use of resources, and alignment with the best international practices.

www.bcb.gov.br

# Inflation target – Central Bank mission (US)

## About the Fed




Transcript (PDF)

The Fed Explained

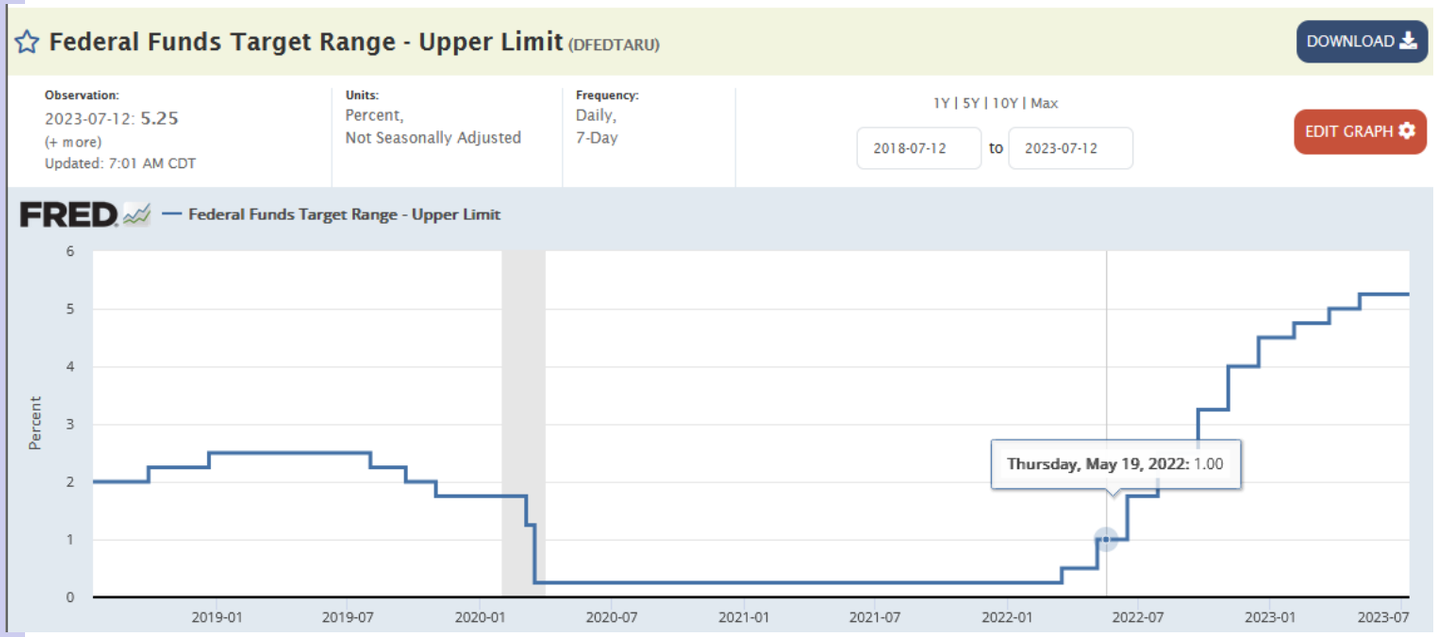
The Federal Reserve System is the central bank of the United States. It performs five general functions to promote the effective operation of the U.S. economy and, more generally, the public interest. The Federal Reserve

- conducts the nation's monetary policy to promote maximum employment, stable prices, and moderate long-term interest rates in the U.S. economy;
- promotes the stability of the financial system and seeks to minimize and contain systemic risks through active monitoring and engagement in the U.S. and abroad;
- promotes the safety and soundness of individual financial institutions and monitors their impact on the financial system as a whole;
- fosters payment and settlement system safety and efficiency through services to the banking industry and the U.S. government that facilitate U.S.-dollar transactions and payments; and
- promotes consumer protection and community development through consumer-focused supervision and examination, research and analysis of emerging consumer issues and trends, community economic development activities, and the administration of consumer laws and regulations.

**… conducts the nation's monetary policy to promote maximum employment, stable prices, and moderate long-term interest rates in the U.S. economy**

https://www.federalreserve.gov/aboutthefed.htm

# US
## Monetary policy and banking system












# Brazilian
Monetary policy and fiscal impacts

Buscar

Valor INTERNATIONAL | Economy

## Public debt rises again under effect of higher interest rates

Indicator totals R$7.5tn, reaches 73.2% of GDP in April, with economic growth preventing greater debt increase

**Por Larissa Garcia, Alex Ribeiro — Brasília, São Paulo**

**31/05/2023 22h24 · Atualizado há um mês**

# Inflation target in Brazil

1. National Monetary Council (CMN) – IPCA target
2. Inflation index – IPCA
3. Interest rate – SELIC
4. Monetary Policy Committee – COPOM
5. COPOM Meetings
6. Expectations – FOCUS
7. Minutes of the Meetings
8. Inflation Report

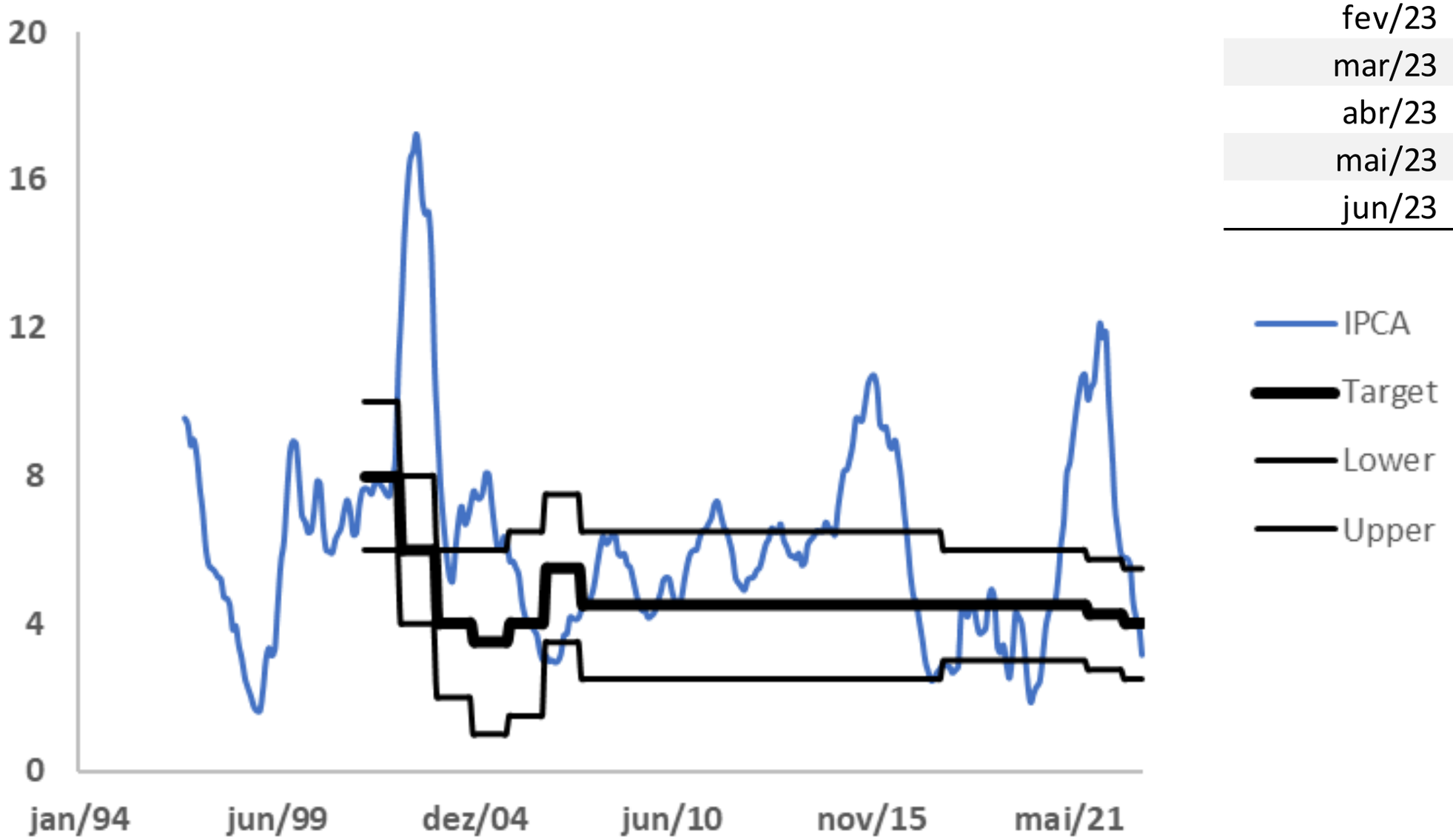
**Let's take a look: www.bcb.gov.br ...**

# Brazilian inflation 12m

2023 – expected 4,95% (Focus report)

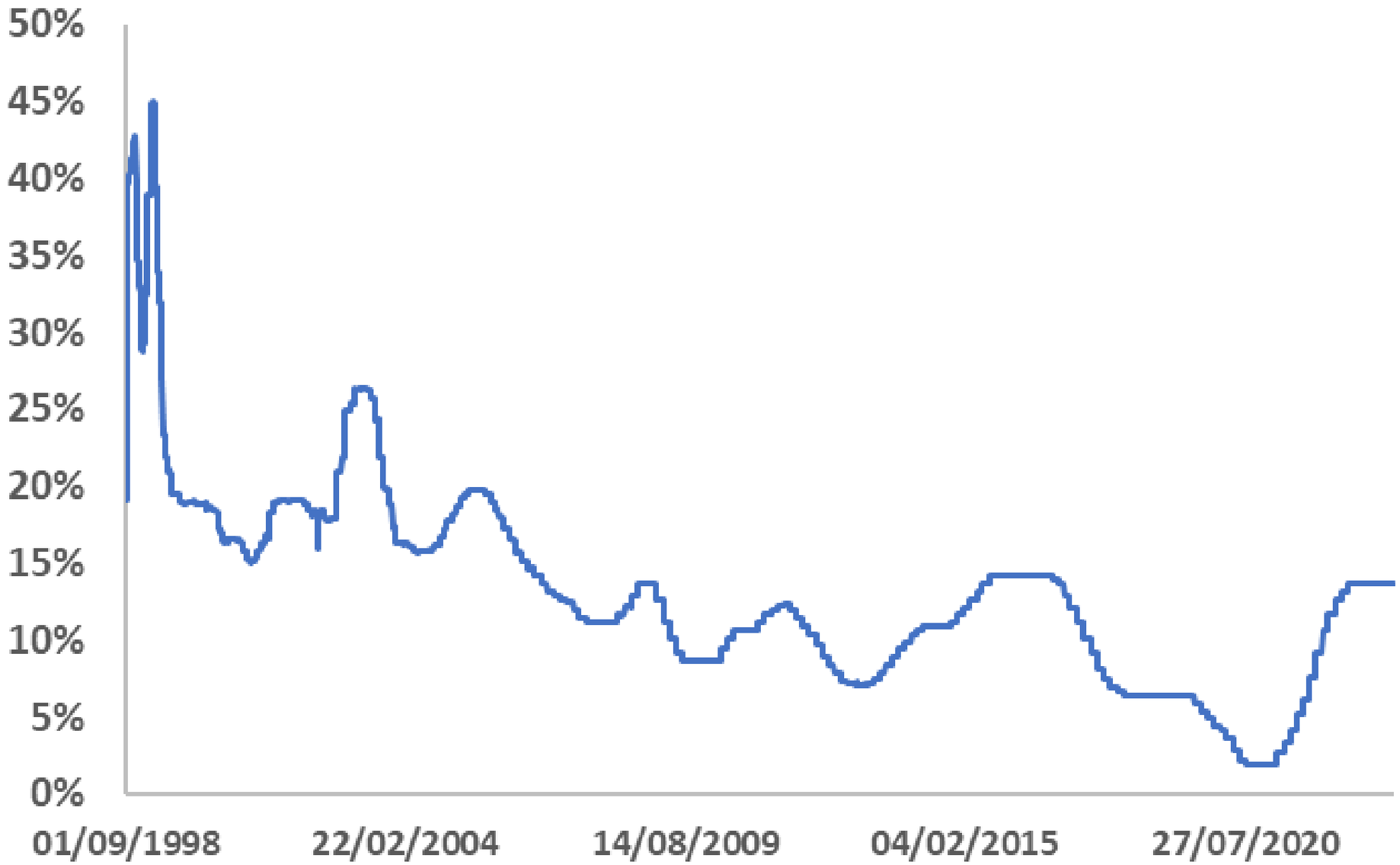
**Target: 3.25% +/- 1.5%**

| | |
|---|---|
| jan/23 | 5.77 |
| fev/23 | 5.6 |
| mar/23 | 4.65 |
| abr/23 | 4.18 |
| mai/23 | 3.94 |
| jun/23 | 3.16 |




www.bcb.gov.br

# Brazilian interest rate 12m



# Questions

1. What will be the next change of interest rate?

Next meetings (2023):

1st and 2nd August

19th and 20th September

31th October and 1st November

12th and 13th December

Selic target July/23: 13,75%
Focus expectations to the end of 2023: 12%

# Group discussion

In your opinion….

1. What are the main drivers of inflation in US?
2. What are the main drivers of inflation in Brazil?
3. In your opinion, how inflation target is working in US?
4. How inflation target is working in Brazil?
5. Is FED independent to run monetary policy?
6. Is Central Bank of Brazil independent to run monetary policy?
7. Should central banks have autonomy to implement monetary policy (to set interest rates in an inflation target program)?

# Summary

- Inflation target
- **Foreign exchange rate regimes**
- International reserves
- The Brazilian strategies to deal with financial crisis

# Foreign Exchange rate regimes

- **Pegged** : the central bank sets a specific foreign exchange rate
- **Currency board**: extreme form of pegged with back of domestic currency in circulation with foreign currency
- **Crawling peg**: a currency with a fixed exchange rate can fluctuate within a band
- **Dirty floating**: the foreign exchange rate fluctuates but the central bank may intervene in the market
- **Free floating**: there is no intervention in the foreign exchange market

# Foreign Exchange rate regimes

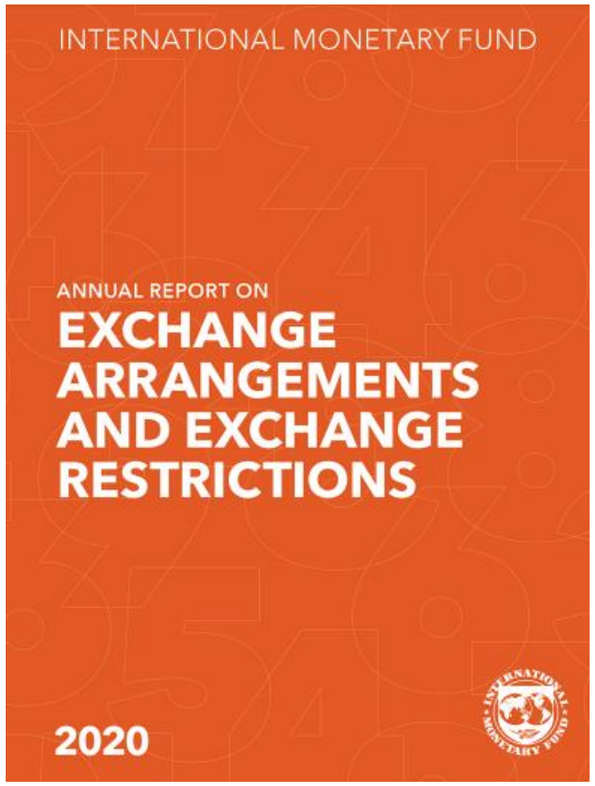



**Table 1. Classification of Exchange Rate Arrangements**

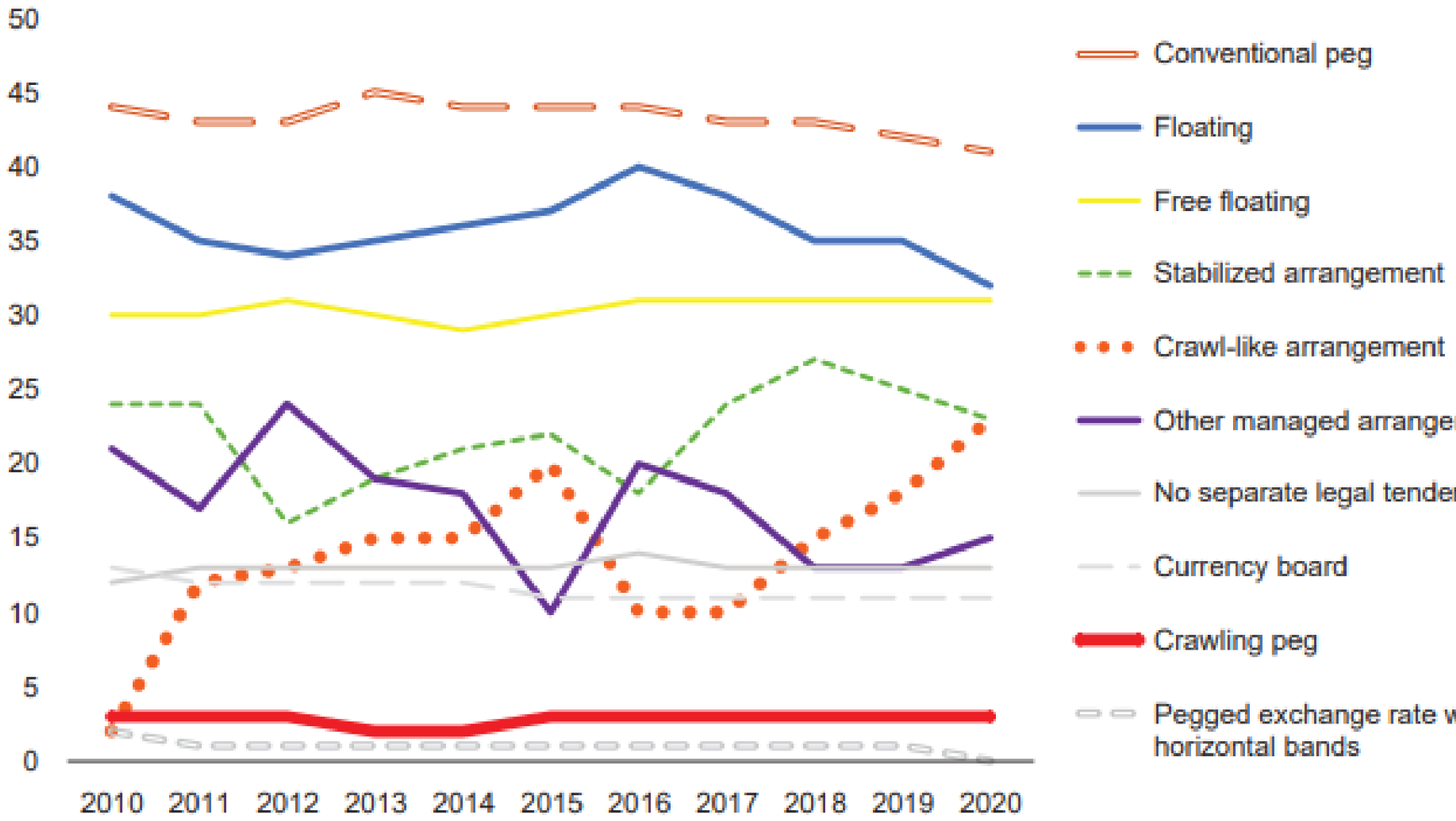
| Type | Categories | | | | |
|---|---|---|---|---|---|
| Hard pegs | Exchange arrangement with no separate legal tender | Currency board arrangement | | | |
| Soft pegs | Conventional pegged arrangement | Pegged exchange rate within horizontal bands | Stabilized arrangement | Crawling peg | Crawl-like arrangement |
| Floating regimes (market-determined rates) | Floating | Free floating | | | |
| Residual | Other managed arrangement | | | | |

Note: This methodology became effective February 2, 2009, and reflects an attempt to provide greater consistency and objectivity of exchange rate classifications across countries and to improve the transparency of the IMF's bilateral and multilateral surveillance in this area. For further details, see IMF Working Paper 09/211.

# Foreign Exchange rate regimes

**Figure 2. Exchange Rate Arrangements, 2010–20**

*(Number of countries as of April 30)*




Source: AREAER database.

# Foreign Exchange rate regimes

**Table 5. Monetary Policy Frameworks and Exchange Rate Anchors, 2012–20**

*(Percent of IMF members as of April 30)*[1]

| Year | US dollar | Euro | Composite | Other currency | Monetary aggregate | Inflation targeting | Other[2] |
|---|---|---|---|---|---|---|---|
| 2012[3] | 22.6 | 14.2 | 6.8 | 4.2 | 15.3 | 16.8 | 20.0 |
| 2013 | 23.0 | 14.1 | 6.8 | 4.2 | 13.6 | 17.8 | 20.4 |
| 2014 | 22.5 | 13.6 | 6.3 | 4.2 | 13.1 | 17.8 | 22.5 |
| 2015 | 22.0 | 13.1 | 6.3 | 4.2 | 13.1 | 18.8 | 22.5 |
| 2016[4] | 20.3 | 13.0 | 4.7 | 4.7 | 12.5 | 19.8 | 25.0 |
| 2017 | 20.3 | 13.0 | 4.7 | 4.7 | 12.5 | 20.8 | 24.0 |
| 2018 | 19.8 | 13.0 | 4.7 | 4.7 | 12.5 | 21.4 | 24.0 |
| 2019 | 19.8 | 13.0 | 4.2 | 4.7 | 13.5 | 21.4 | 23.4 |
| 2020[5] | 19.8 | 13.0 | 4.2 | 4.7 | 11.5 | 22.4 | 24.5 |

Source: AREAER database.

[1] Includes 189 member countries and the following territories: Aruba, Curaçao and Sint Maarten (all in the Kingdom of the Netherlands: information for Curaçao and Sint Maarten is reported together as they have a common central bank) and Hong Kong SAR (People's Republic of China).

[2] Includes countries that have no explicitly stated nominal anchor but instead monitor various indicators in conducting monetary policy.

[3] Does not include South Sudan, which became an IMF member on April 18, 2012.

[4] Includes Nauru, which became an IMF member on April 12, 2016.

[5] Does not include Andorra, which became an IMF member on October 16, 2020.

# Foreign exchange rate interventions

Central banks intervene in the foreign exchange market buying or selling foreign currency. They may intervene to build reserves or to dampen excessive market volatility, but they may also intervene to fight depreciation pressure.

IMF (2021). Annual Report on Exchange Arrangements and Exchange restrictions

# Foreign exchange rate interventions

## Techniques

- **Spot foreign exchange market:** by contacting market participants or through auctions
- **Forward, swaps or options markets**
- **Verbal interventions**
- **Others**: Capital controls and interest rate changes may also influence foreign exchange rates

# Foreign exchange rate interventions

## Articles IV, VIII and XIV:

## Rules for currencies, restrictions, foreign-held balances and reserve assets



# Foreign exchange rate interventions

## Restrictions

- Trade  (import and export tariffs)
- Capital controls (inflow and outflow for investments)
- Transactions (credit card payments)

# Foreign exchange rate interventions

## Restrictions on transactions

The IMF staff report for the 2019 Article IV Consultation with Brazil states that, as of June 26, 2019, the tax on financial transactions (Imposto sobre Operações Financeiras, IOF) of 6.38% on exchange transactions carried out by credit card, debit card, and traveler's checks (including cash withdrawals) companies in order to fulfill their payment obligations for purchases of goods and services abroad by their customers gives rise to a multiple currency practice (MCP) subject to IMF jurisdiction under Article VIII, Sections 2(a) and 3. In January 2008, the IOF for these exchange transactions was raised to 2.38% and then further increased to 6.38% in March 2011. The scope of operations was expanded to other foreign exchange transactions than with credit cards in December 2013. (Country Report No. 19/242)

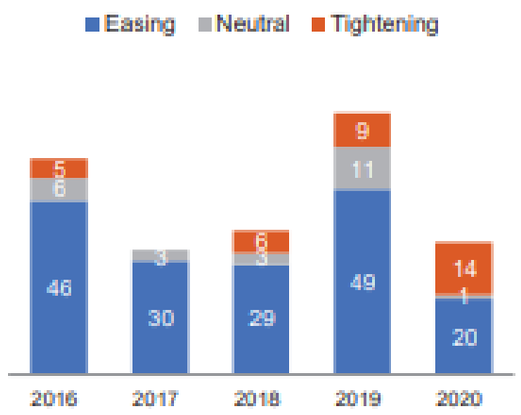
IMF (2021). Annual Report on Exchange Arrangements and Exchange restrictions

# Foreign exchange rate interventions

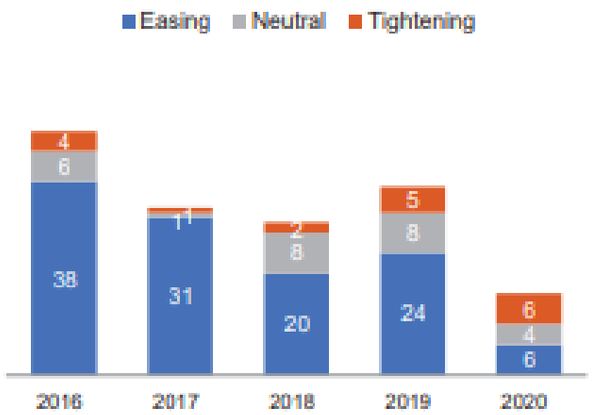
**Figure 8. Account Transactions**

*(Number of changes, by measure)*

1. Resident accounts: in 2019, liberalization actions largely outnumbered tightening and neutral actions, in line with past trends. The trend continued in 2020, although at a slower pace.




2. Nonresident accounts: easing measures continued to dominate in 2019, but the trend was more balanced in 2020.




Source: AREAER database.
Note: The position date for 2020 varies by country and is less than the full year.

IMF (2021). Annual Report on Exchange Arrangements and Exchange restrictions

## Foreign exchange rate interventions

- Sterilized interventions: if the central bank buys (sells) dollar it sells (buys) domestic bonds to sterilize monetary expansion;

- Non-Sterilized interventions: the central bank buys or sells dollars without monetary impact concerns.

IMF (2021). Annual Report on Exchange Arrangements and Exchange restrictions

# Foreign exchange rate – theory

## Covered interest rate parity

It is a non-arbitrage condition in which the relationship between spot and forward currency values of two countries **is equal** by the difference between interest rates:

$$F = S * \frac{1 + i_d}{1 + i_f}$$

$F$: forward foreign exchange rate (forward rate today)

$S$: spot foreign exchange rate

$i_d$: domestic interest rate

$i_f$: foreign interest rate

# Foreign exchange rate – theory

## Uncovered interest rate parity (UIP)

According to UIP, the relative change in foreign exchange rate of two currencies **will equal** the difference in interest rates between the two countries in the same period.

$$F_u = S * \frac{1 + i_d}{1 + i_f}$$

$F_u$: forward foreign exchange rate (spot rate in the future)

$S$: spot foreign exchange rate

$i_d$: domestic interest rate

$i_f$: foreign interest rate

# Foreign exchange rate – theory

## Purchasing power parity (PPP)

- Law of one price
- No transaction costs
- No trade barriers

# Questions

1. How US monetary policy affect foreign exchange rates in other countries?
2. What are the effects of foreign exchange rate in country competitiveness and financial stability?
3. Currency interventions are effective in managing the exchange rate?

# The Trilemma in Emerging Market Economies

The trilemma postulates that countries face a trade-off among the objectives of exchange rate stability, free capital mobility, and independent monetary policy.

Obstfeld, M. Ostry, J., Qureshi, M. (2017). A Tie that Binds: Revisiting the Trilemma in Emerging Market Economies. IMF Working Papers N.17/130

# The Trilemma in Emerging Market Economies

It is not possible to have at the same time monetary policy independence, free capital mobility and foreign exchange rate stability. Central banks have to choose only two of the three objectives.

Obstfeld, M. Ostry, J., Qureshi, M. (2017). A Tie that Binds: Revisiting the Trilemma in Emerging Market Economies. IMF Working Papers N.17/130

# The Trilemma in Emerging Market Economies

If a country chooses exchange rate stability and free capital mobility, it must give up monetary policy autonomy; conversely, an independent monetary policy in the presence of free capital flows is possible through exchange rate flexibility. Throughout history, countries have attempted to navigate the trilemma by choosing different policy configurations.

Obstfeld, M. Ostry, J., Qureshi, M. (2017). A Tie that Binds: Revisiting the Trilemma in Emerging Market Economies. IMF Working Papers N.17/130

# The Trilemma in Emerging Market Economies

Floating foreign exchange rate regime implies more monetary independence (short term interest rates);

However, the capacity of central banks to set policy rates under flexible exchange rate regimes does not ensure their ability to influence domestic monetary and financial conditions.

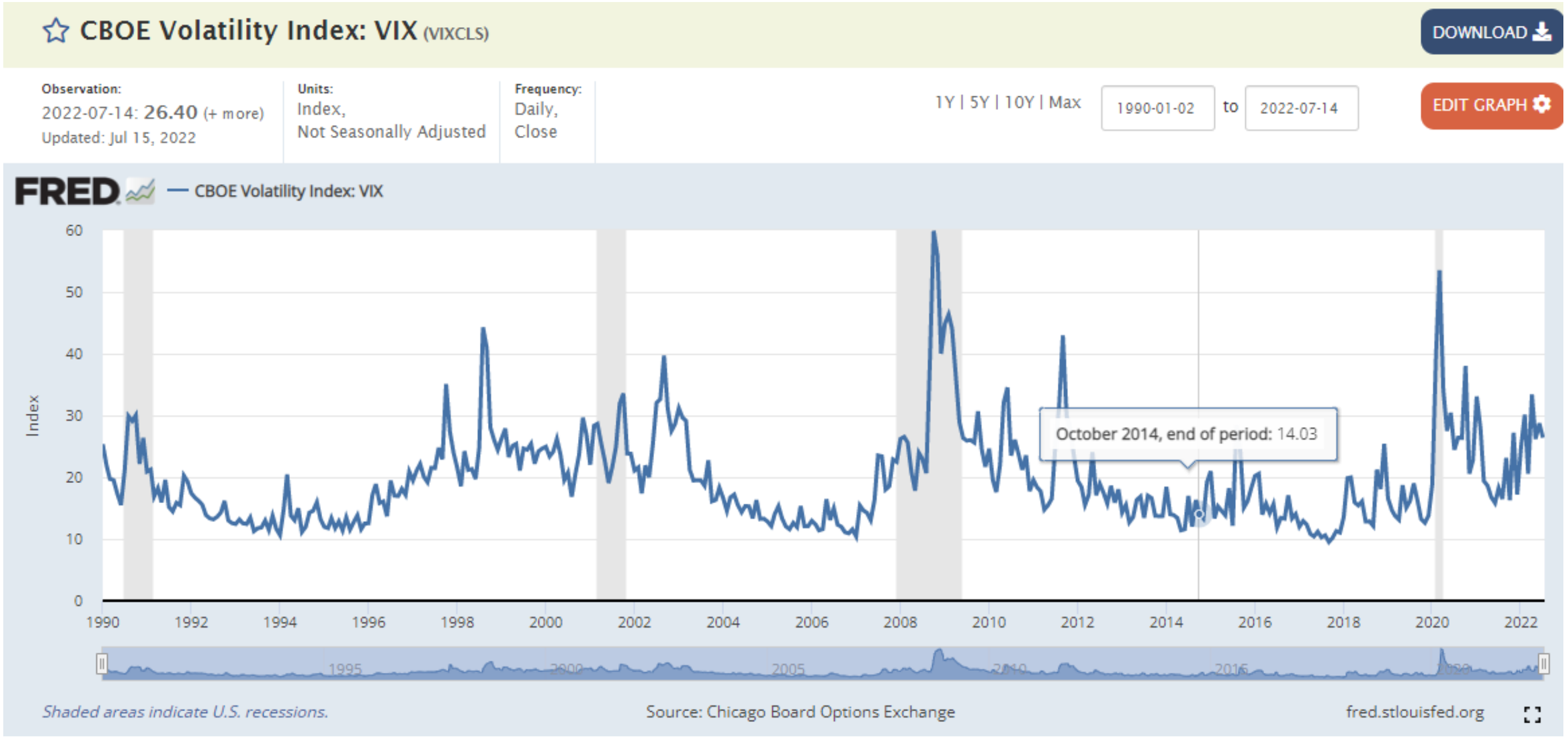
Obstfeld, M. Ostry, J., Qureshi, M. (2017). A Tie that Binds: Revisiting the Trilemma in Emerging Market Economies. IMF Working Papers N.17/130

# Capital flows

- Foreign direct investments
- Portfolio investments

Volatility of capital flows does matter for emerging market economies

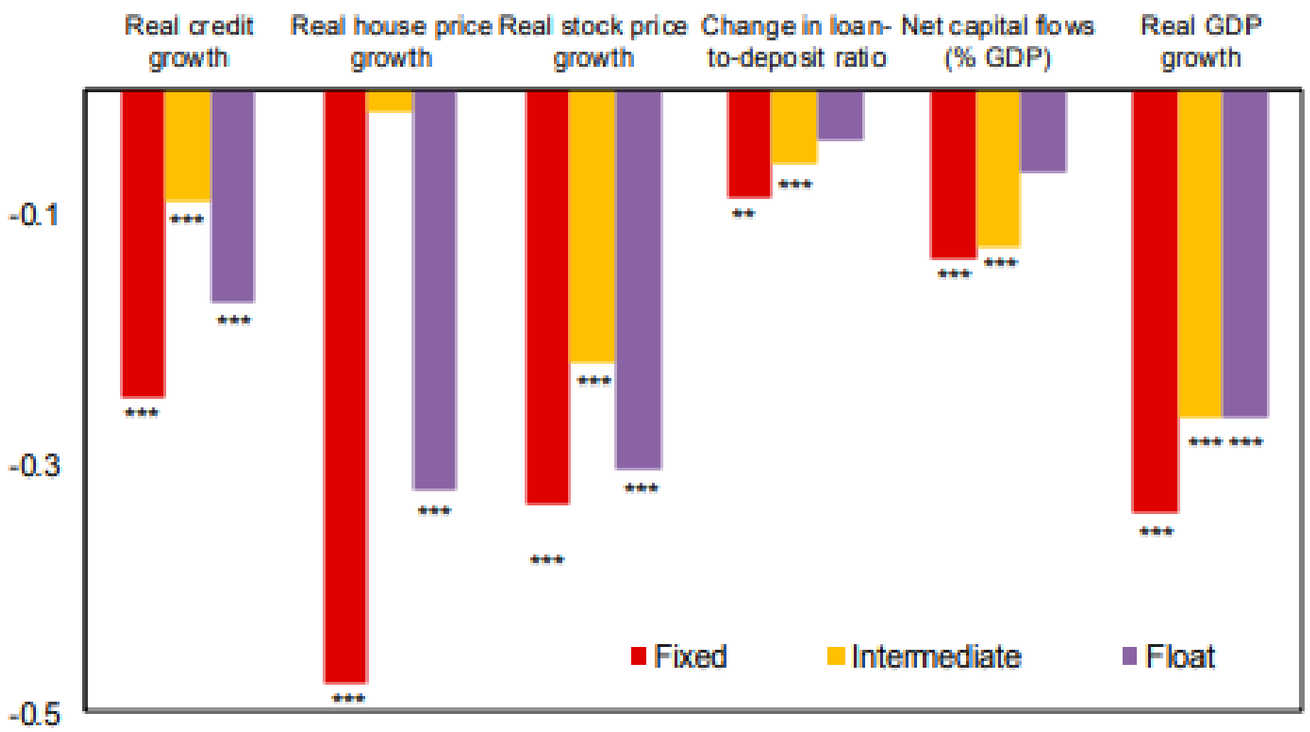
# VIX: a proxy for capital flows in emerging markets



# VIX: a proxy for capital flows in emerging markets

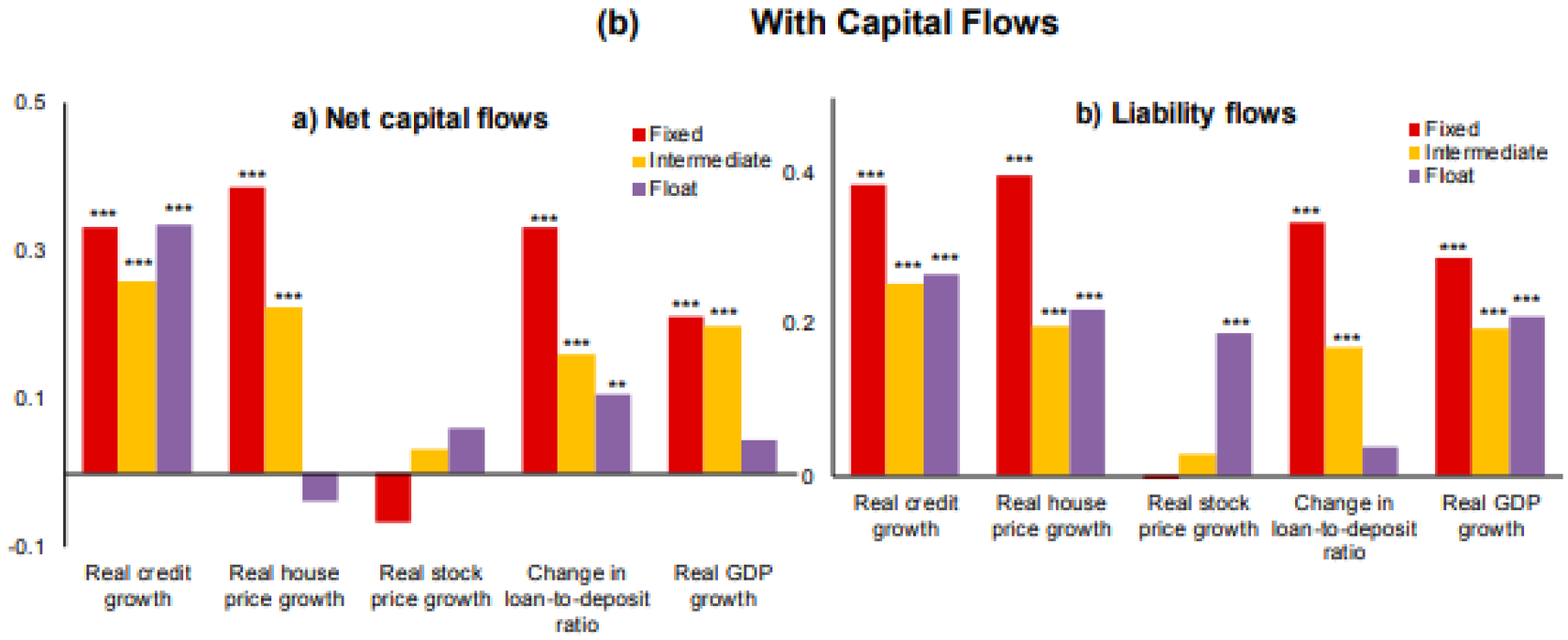
**Figure 5. Correlation of Financial and Macroeconomic Variables in EMEs**

**(a) With Global Investor Risk Aversion**




Obstfeld, M. Ostry, J., Qureshi, M. (2017). A Tie that Binds: Revisiting the Trilemma in Emerging Market Economies. IMF Working Papers N.17/130

# Correlations between Capital flows and financial and economic variables in emerging markets




Source: Authors' calculations.
Note: Panel [a] shows the unconditional correlation across countries between the (log) VXO index and three-quarter moving average of real domestic private sector credit growth, real house price growth, real stock price growth, change in loan-to-deposit ratio, net capital flows to GDP, and real GDP growth. Panel [b] shows the unconditional correlation between three-quarter moving average of net and liability flows, and real domestic private sector credit growth, real house price growth, real stock price growth, change in loan-to-deposit ratio, net capital flows to GDP, and real GDP growth. *,**, *** indicate statistical significance at the 10, 5, and 1 percent levels, respectively.

Obstfeld, M. Ostry, J., Qureshi, M. (2017). A Tie that Binds: Revisiting the Trilemma in Emerging Market Economies. IMF Working Papers N.17/130

# Questions

1. Why Greece’s debt crisis was a big deal to euro?
2. Why Argentina’s currency board crisis was difficult to solve?
3. Why Brazil had to change its foreign exchange rate regime in 1999?

# Summary

- Inflation target
- Foreign exchange rate regimes
- **International reserves**
- The Brazilian strategies to deal with financial crisis

# What are international reserves?

- International reserves consist of official public sector foreign assets that are readily available to and controlled by the monetary authorities.

Source: International Monetary Fund: https://www.imf.org/external/np/mae/ferm/eng/

# Questions

1. Why countries should hold international reserves?
2. What should be the optimal level of international reserves in a country (the adequacy problem)?
3. Emerging market countries should hold more international reserves than developed countries?

# Why countries should hold international reserve assets?

- Traditionally the rationale behind holding reserves associates them with precaution and transaction considerations.
- Mercantilist approach for foreign exchange rate
- Monetary policy considerations
- Wealth management

Source: Silva Jr. (2011). The Self-insurance Role of International Reserves and the 2008-2010 Crisis. Working Paper Series n. 256, Banco Central do Brasil.

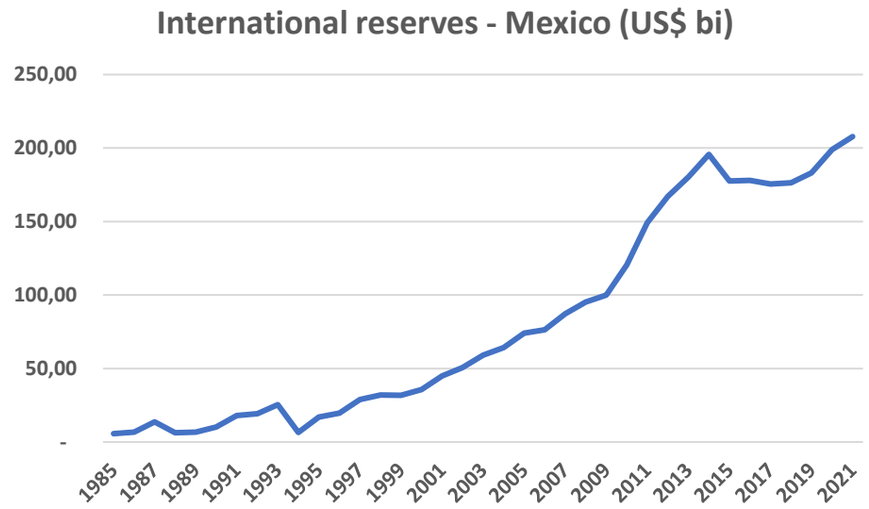
# Optimal level of international reserves

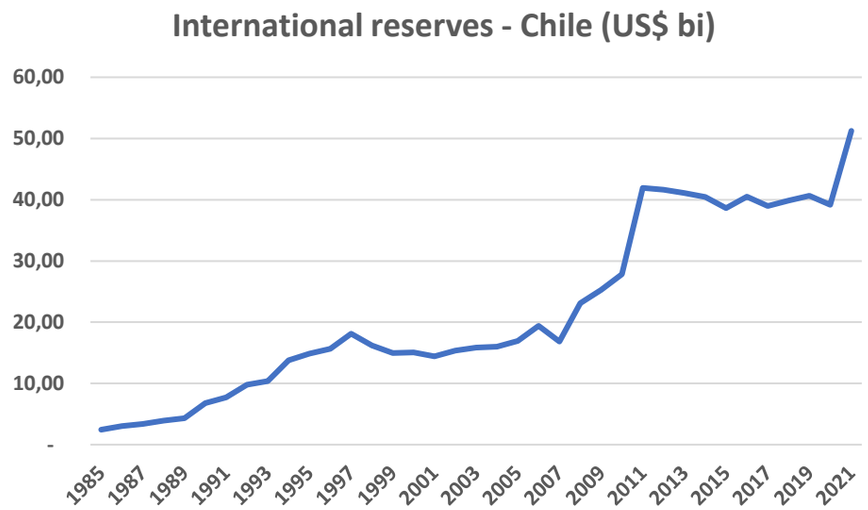
- An adequate level of international reserves was previously considered to be the amount needed to cover at least three months of imports;
- The ratio of reserves to short-term external debt is also a guide to reserve adequacy;
- Cost-benefit approach

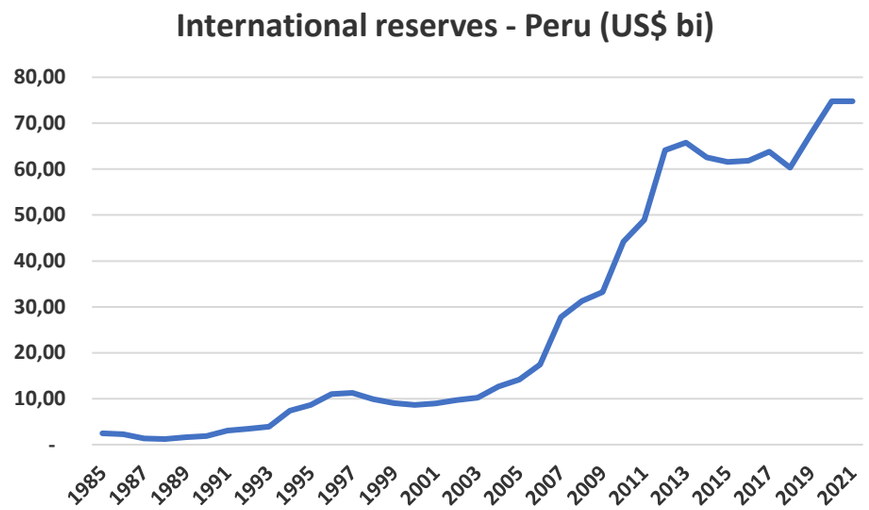
Source: Silva Jr. (2011). The Self-insurance Role of International Reserves and the 2008-2010 Crisis. Working Paper Series n. 256, Banco Central do Brasil.

# International reserves – Latin America

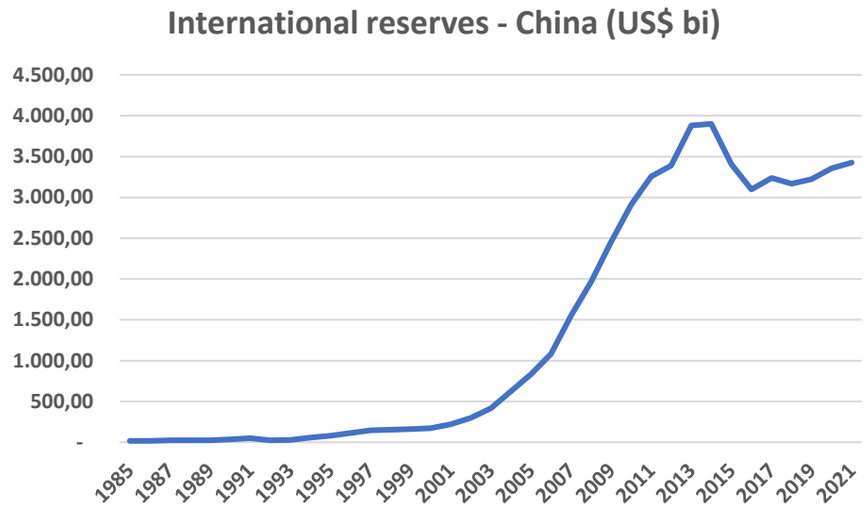


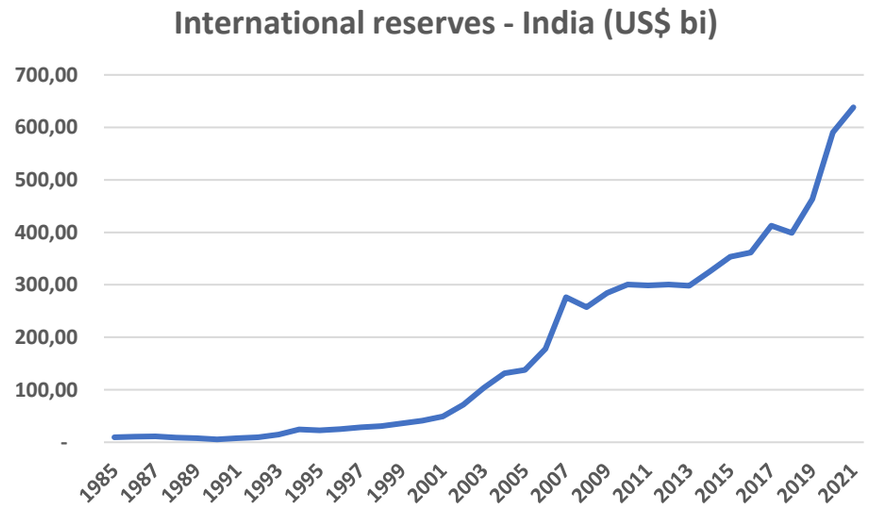



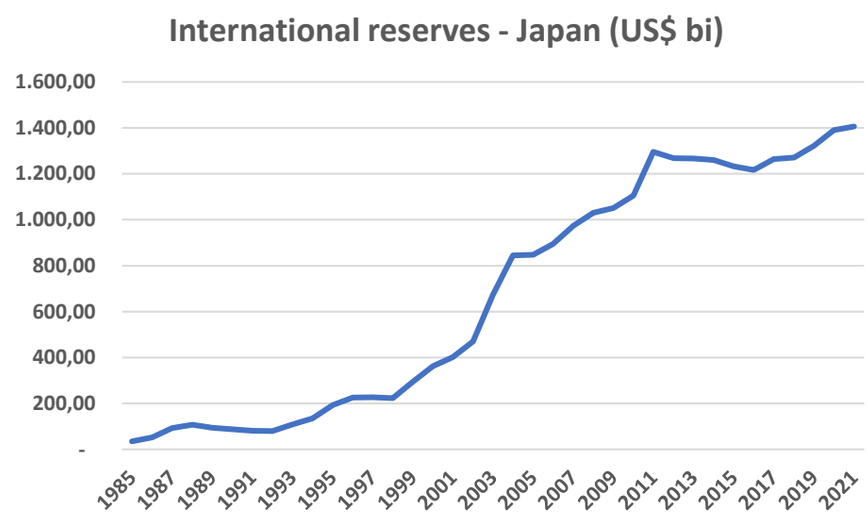



# International reserves – Asia





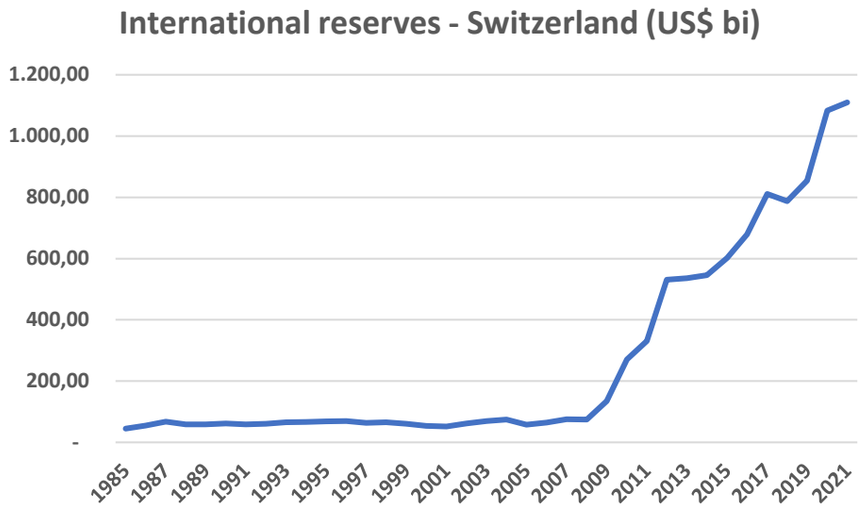

# International reserves – Europe



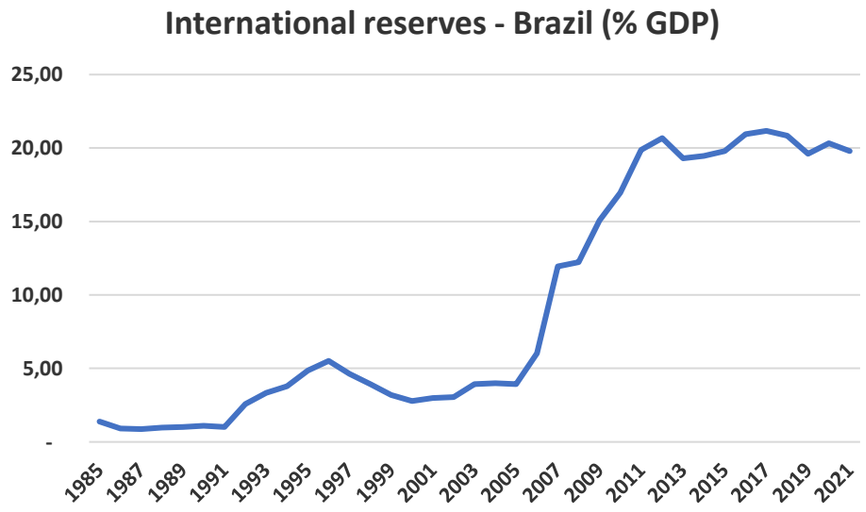


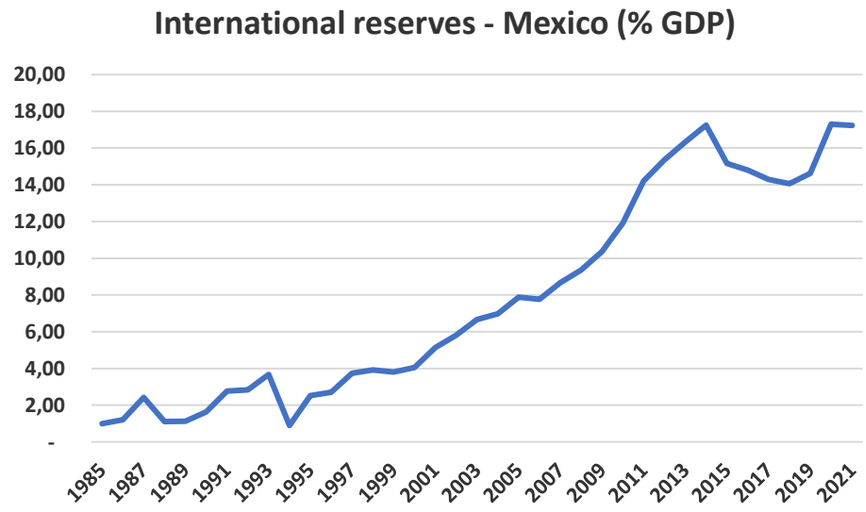


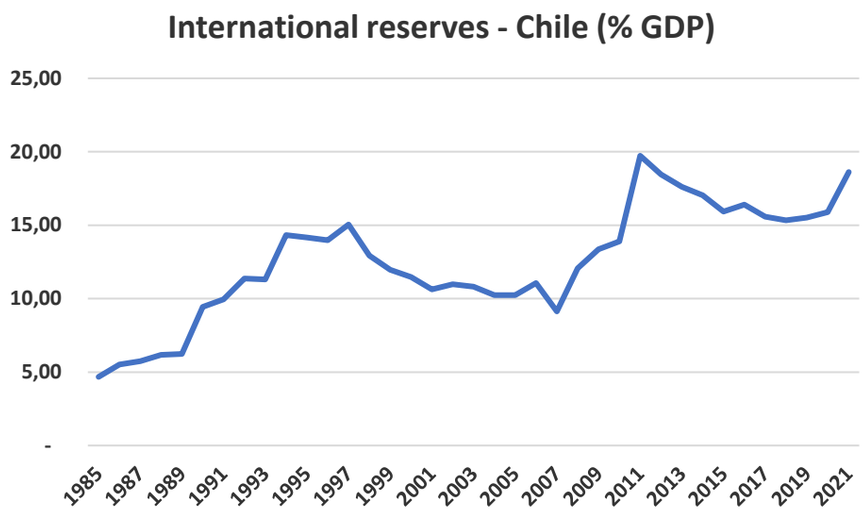


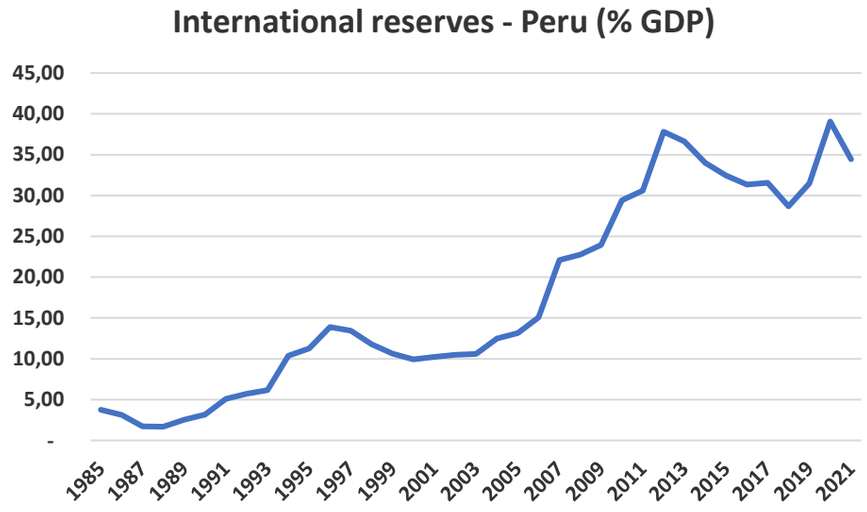



# International reserves – Latin America (% GDP)



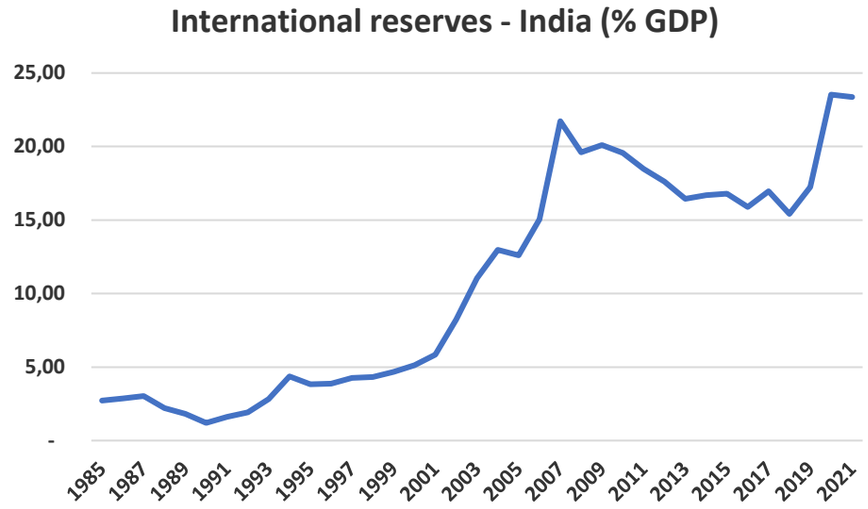



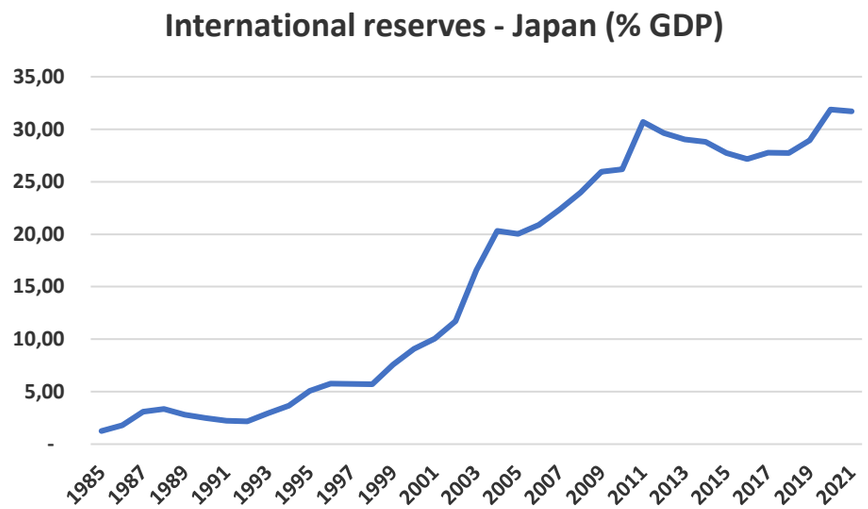



# International reserves – Asia (% GDP)

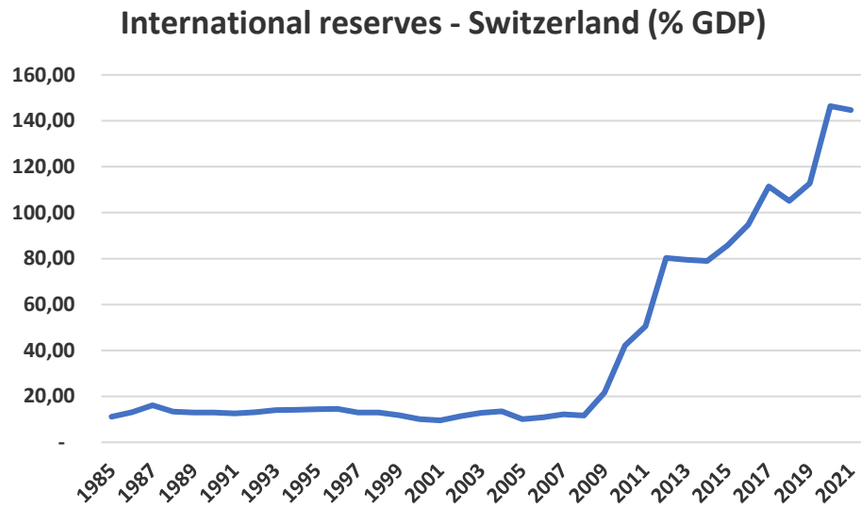




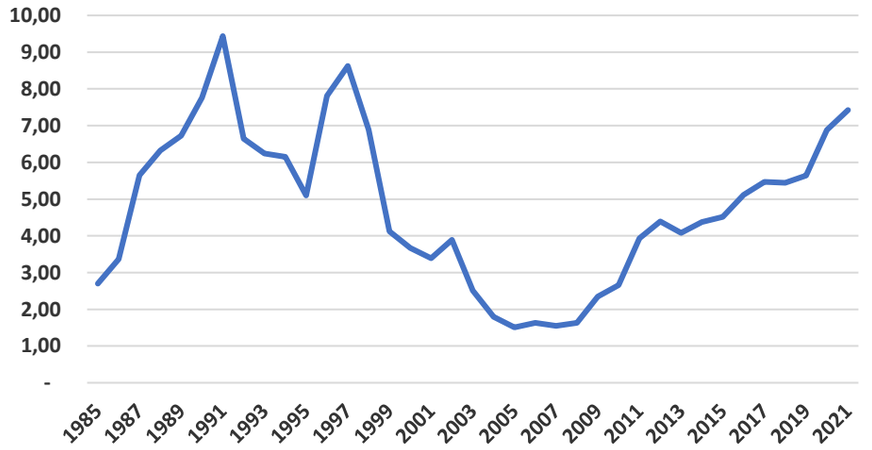

# International reserves – Europe (% GDP)









# Optimal level of international reserves

- Costs of holding international reserves: interest rate differential and foreign exchange rate change

- Benefits: avoiding GDP fall (crises)

Source: Silva Jr. (2011). The Self-insurance Role of International Reserves and the 2008-2010 Crisis. Working Paper Series n. 256, Banco Central do Brasil.

# Optimal level of international reserves

- There is no standard rule for the definition of an “optimal level” of international reserves and several assumptions underlie the rationale behind holding reserves. There are various theoretical approaches, but no standard for the evaluation of the performance of “optimal level” models, and their parameters are difficult to estimate.

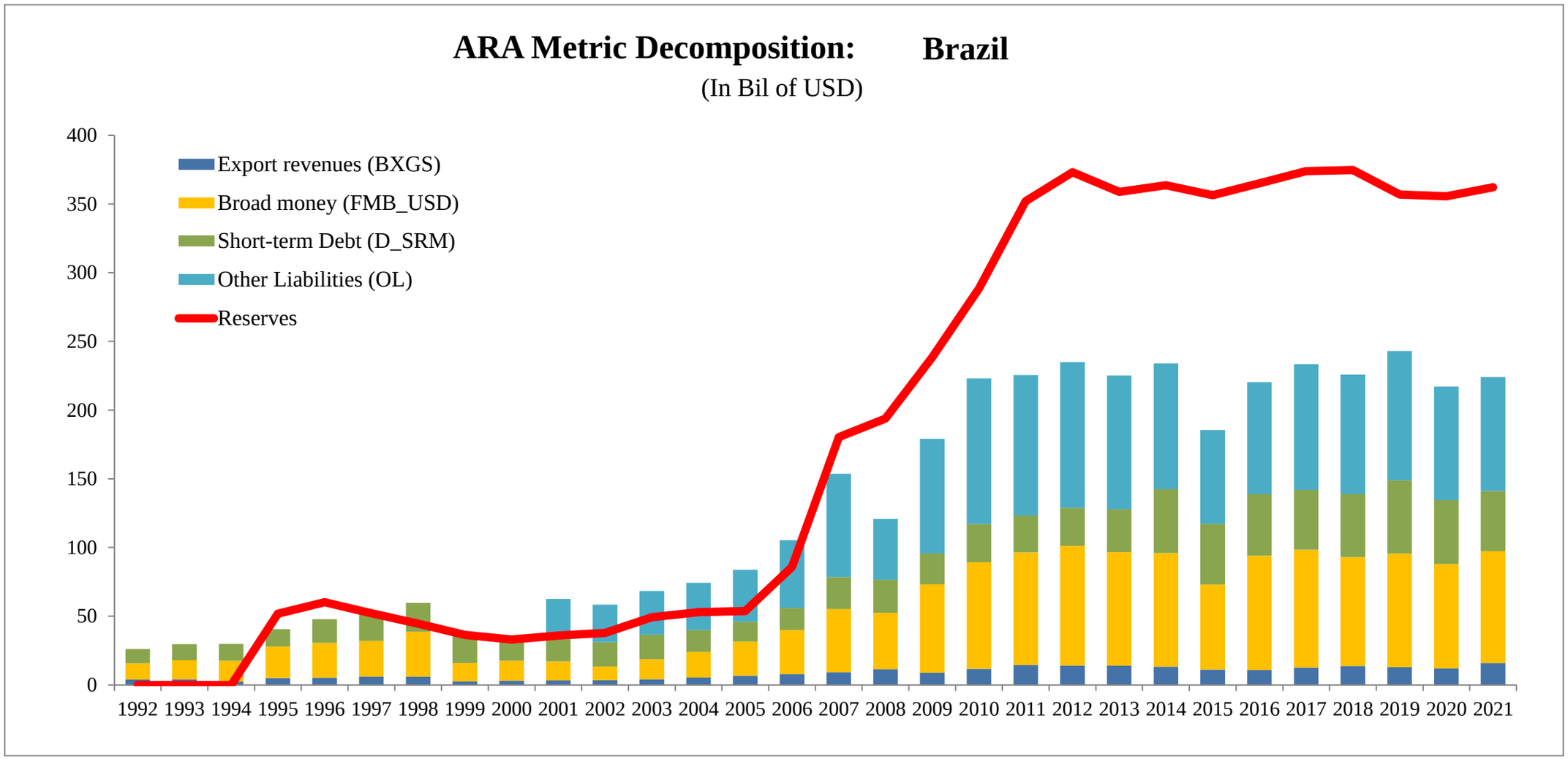
Source: Silva Jr. (2011). The Self-insurance Role of International Reserves and the 2008-2010 Crisis. Working Paper Series n. 256, Banco Central do Brasil.

# Assessing reserve adequacy – ARA (IMF)

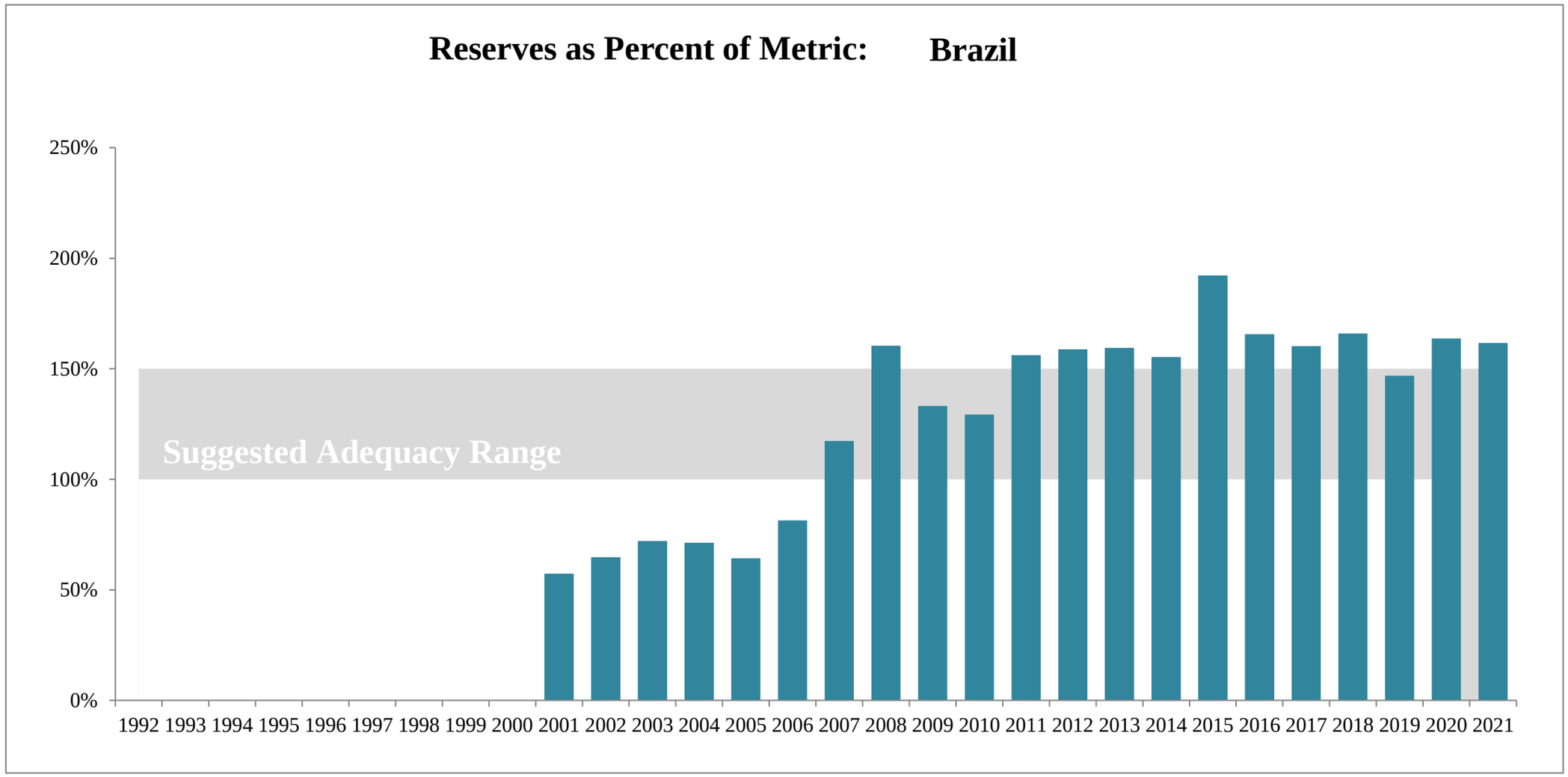
- IMF uses an approach based on short term debt, other liabilities, broad money and exports;
- Data is available in:

https://www.imf.org/external/datamapper/datasets/ARA

# Assessing reserve adequacy – ARA (IMF)



# Assessing reserve adequacy – ARA (IMF)



# Stylized facts

- Vast literature about international reserves holdings

- Developed countries hold less reserves but there are some interesting cases (see Switzerland)

- Asian countries hold more reserves

# Empirical evidence in the Global Financial Crisis

$$g_i - \bar{g}_i = \theta_1 + \theta_2(market_i + debt_i) + \theta_3(reserves_i - E[reserves_i]) + \epsilon_i$$

Table 3 – Average GDP fall in three years as a function of reserves and capital market size

| Variable | Coefficient | Std. Error | p-Value |
|---|---|---|---|
| Constant | -0.0290 | 0.0038 | 0.0000 |
| Market+Debt | -0.0011 | 0.0004 | 0.0052 |
| Reserves-E(Reserves) | 0.0723 | 0.0207 | 0.0008 |

$R^2$: 0.296

White correction for heteroscedasticity.

Source: Silva Jr. (2011). The Self-insurance Role of International Reserves and the 2008-2010 Crisis. Working Paper Series n. 256, Banco Central do Brasil.

# Empirical evidence in the Global Financial Crisis

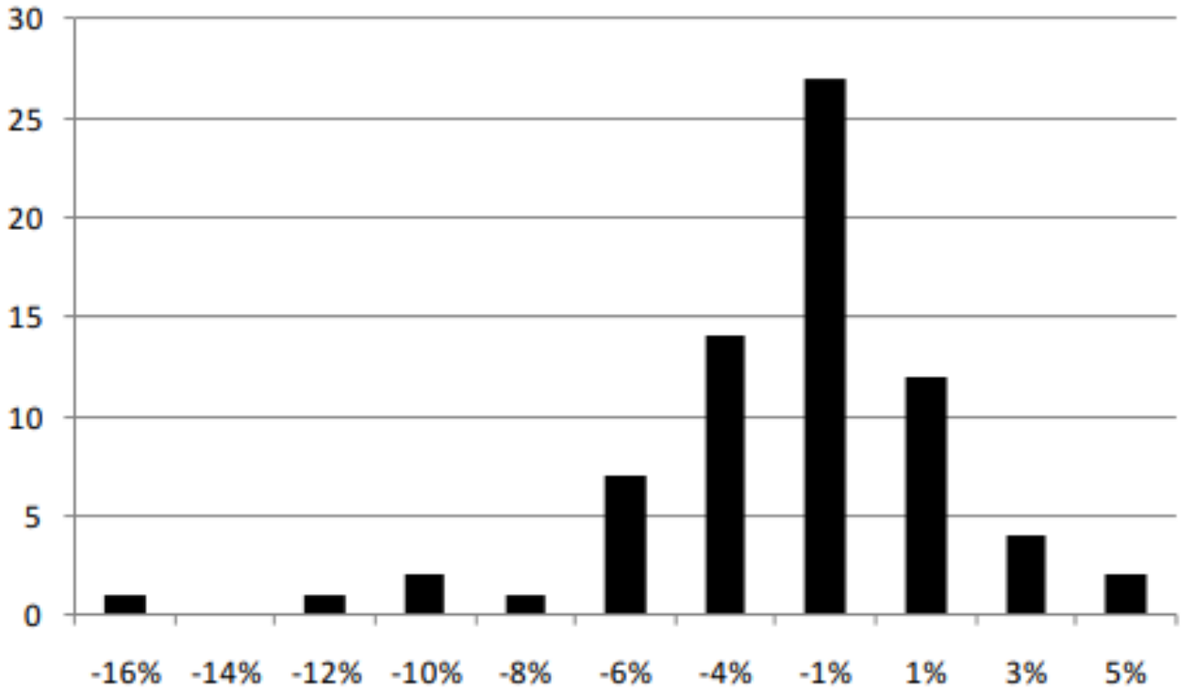



Figure 2 – Histogram of annual average GDP growth minus expected growth in the period 2008-2010

Source: Silva Jr. (2011). The Self-insurance Role of International Reserves and the 2008-2010 Crisis. Working Paper Series n. 256, Banco Central do Brasil.

# Empirical evidence in the Global Financial Crisis

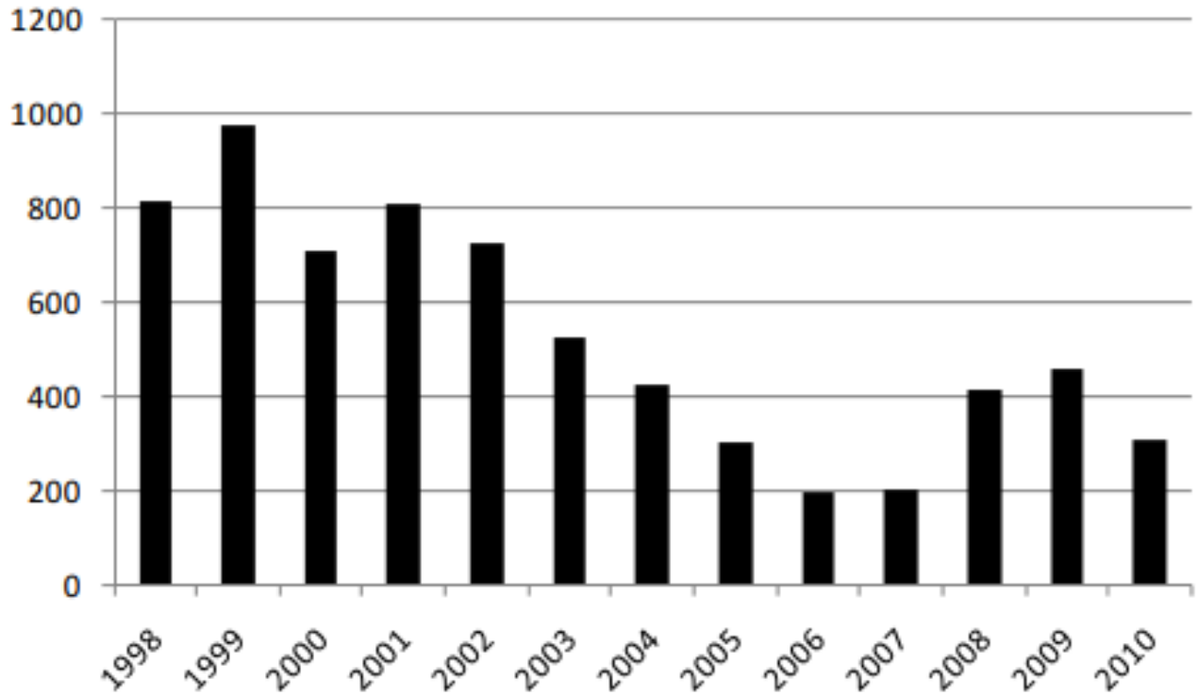



Figure 3 – Embi Global stripped spread (basis points)

Source: Silva Jr. (2011). The Self-insurance Role of International Reserves and the 2008-2010 Crisis. Working Paper Series n. 256, Banco Central do Brasil.

# Questions

1. How uncovered interest rate parity may affect costs of holding international reserves?
2. What would be the impact to the Brazilian international reserves of a genuine free-float regime?

# Summary

- Inflation target
- Foreign exchange rate regimes
- International reserves
- **The Brazilian strategies to deal with financial crisis**

# The Brazilian strategies during crises

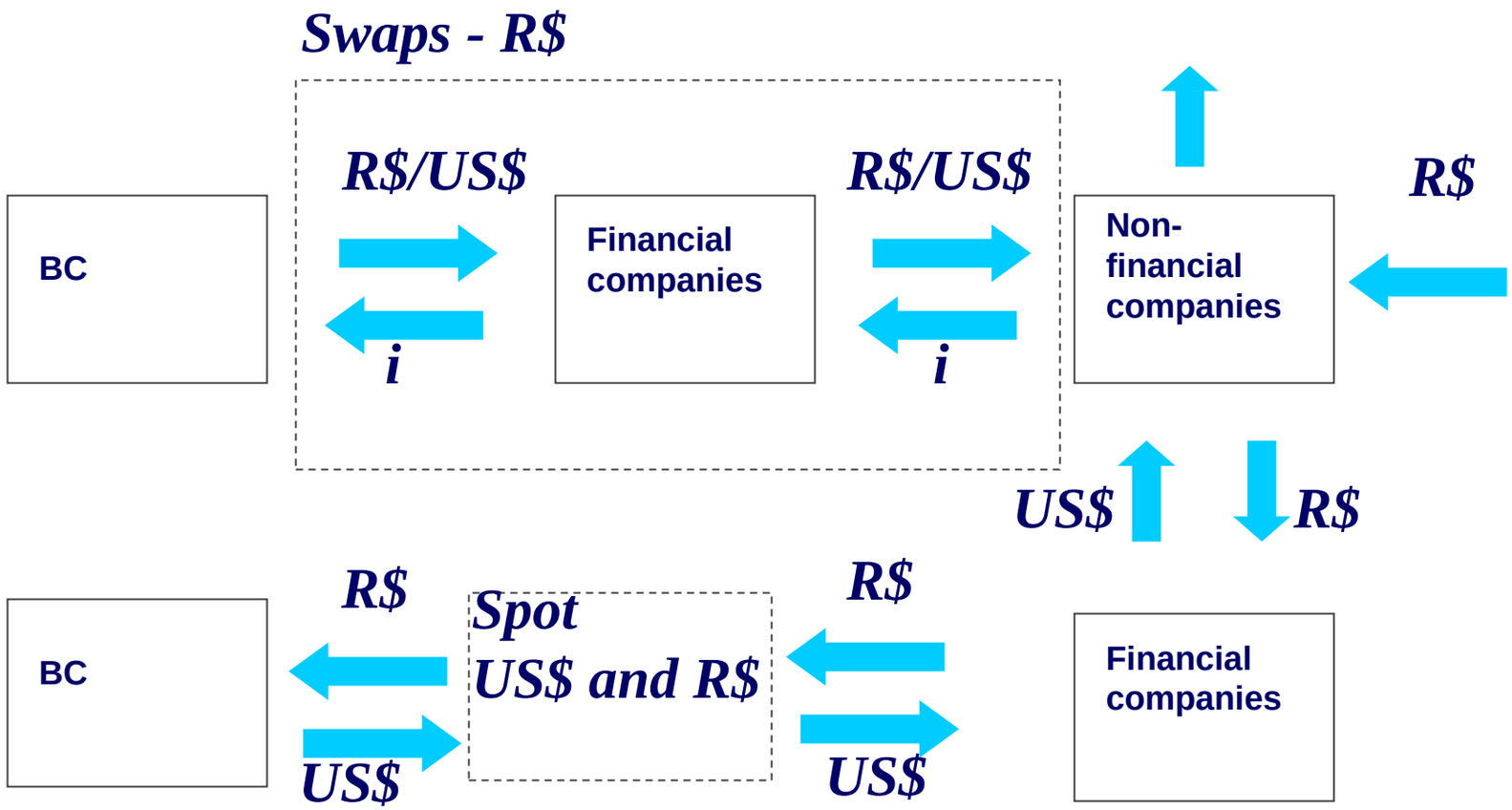
There is a huge debate on the efficacy of foreign exchange rate interventions but even in a floating foreign exchange rate regime central banks usually intervene!

# The Brazilian strategies during crises

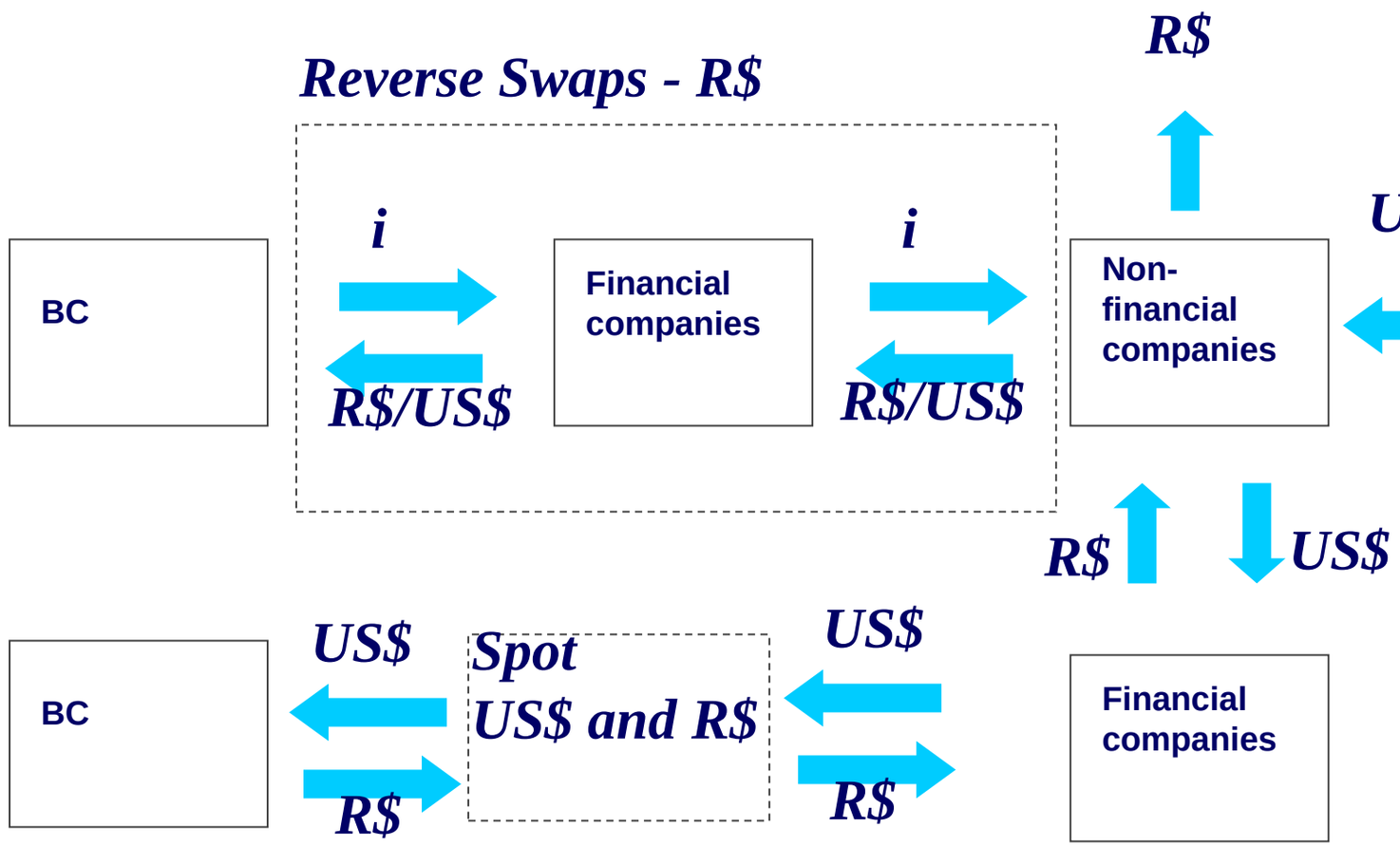
Among other strategies, Brazil relies on interventions in:

- the spot market with international reserves
- swaps of real x dollar (similar to a non-delivery forward)

# Foreign exchange rate swaps



# Foreign exchange rate swaps



# Exchange rate stochastic process

$$Y_t = y + \int_0^t p \cdot ds - \int_0^t b \cdot r_s ds + \sigma B_t - a_1 \sum_{j:\theta_j \le t} \xi_{1,j} - a_2 \sum_{j:\theta_j \le t} \xi_{2,j}$$

| | |
|---|---|
| $Y_t$ | is the exchange rate stochastic process starting at *y*; |
| $p$ | is the exogenous economic pressure on the exchange rate; |
| $b.r_s$ | is the influence of interest rate differential ($r_s$) in the exchange rate |
| $\sigma$ | is a positive constant; |
| $B_t$ | is the Brownian motion; |
| $a_1.\xi_1$ | is the influence of interventions in the exchange rate by selling ($\xi>0$) or buying ($\xi<0$); |
| $a_2.\xi_2$ | is the influence of interventions in the exchange rate by selling ($\xi>0$) or buying ($\xi<0$) swaps. |

# Exchange rate stochastic process

**Cost function**

$$J^{w}(s,y) = E^{(s,y)}\left[\int_{s}^{T} e^{-\rho t}\left(k(Y_t - m)^2 + \beta\left(r_t - \bar{r}\right)^2\right)dt + \sum_{i=1}^{2}\sum_{j:\theta_j \le T} e^{-\rho\theta_j}\left(c_i + \lambda_i \xi_{ij}^2\right)\right]$$

Proxy of foreign exchange rate deviations

Continuous effect of interest rate on foreign exchange rate

Impulse control:

- international reserves
- swaps

# Exchange rate stochastic process – model identification

## Brazilian reduction in international reserves in previous crises

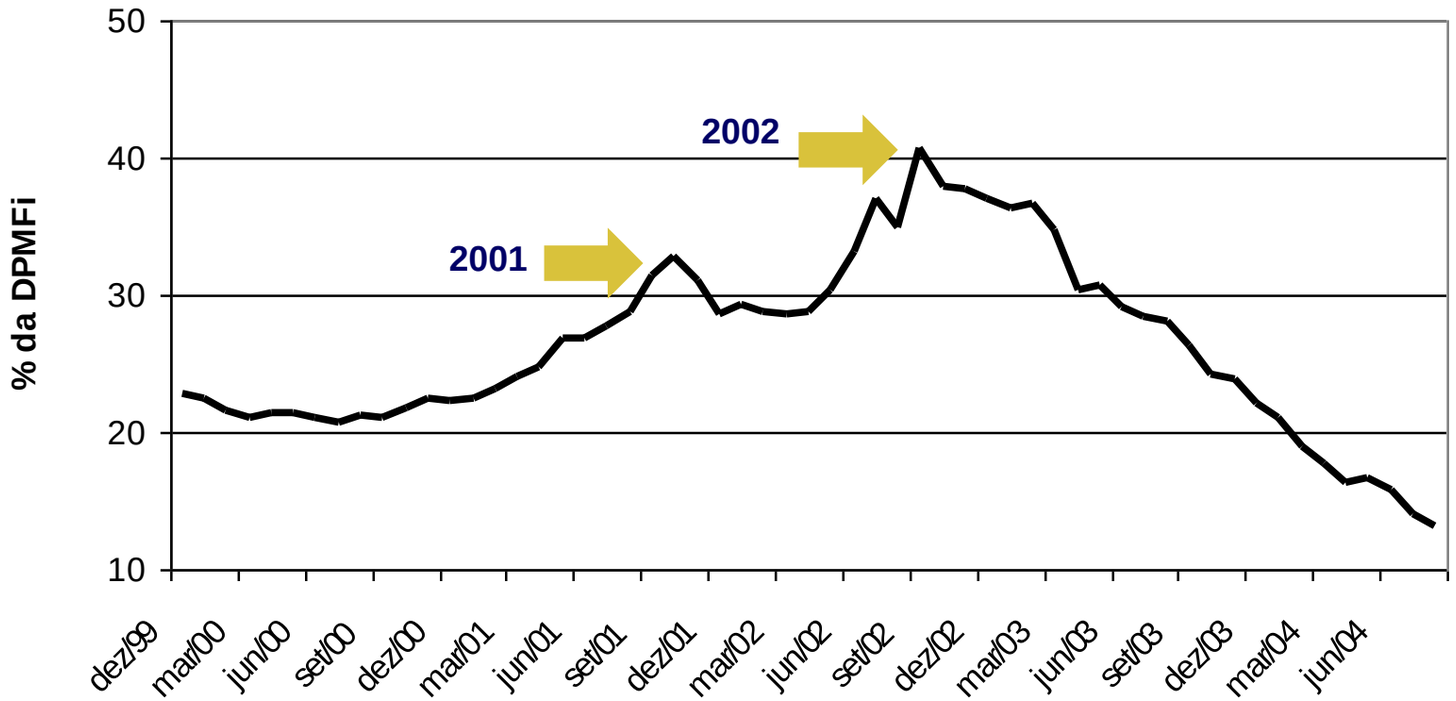
| Crisis | Period | US$ |
|---|---|---|
| Mexico | oct/94-apr/95 | 11.568 |
| Asia | aug/97-nov/97 | 11.021 |
| Russia | apr/98-nov/98 | 33.467 |
| Brazil - regime | dec/98-mar/99 | 10.708 |
| Argentina | sep/01-may/02 | 7.165 |
| Brasil - election | jun/02-nov/02 | 6.407 |

# Exchange rate stochastic process – model identification

| | 2001 | 2002 |
|---|---|---|
| Interventions with international reserves(US$ bi) | 8.2 | 7.4 |
| Interventions with swaps or bonds (US$ bi) | 3.7 | 1.7 |
| Interest rate differential (%) | 16.4 | 18.1 |

# Exchange rate stochastic process – model identification

## % of FX short position on Internal Public Debt



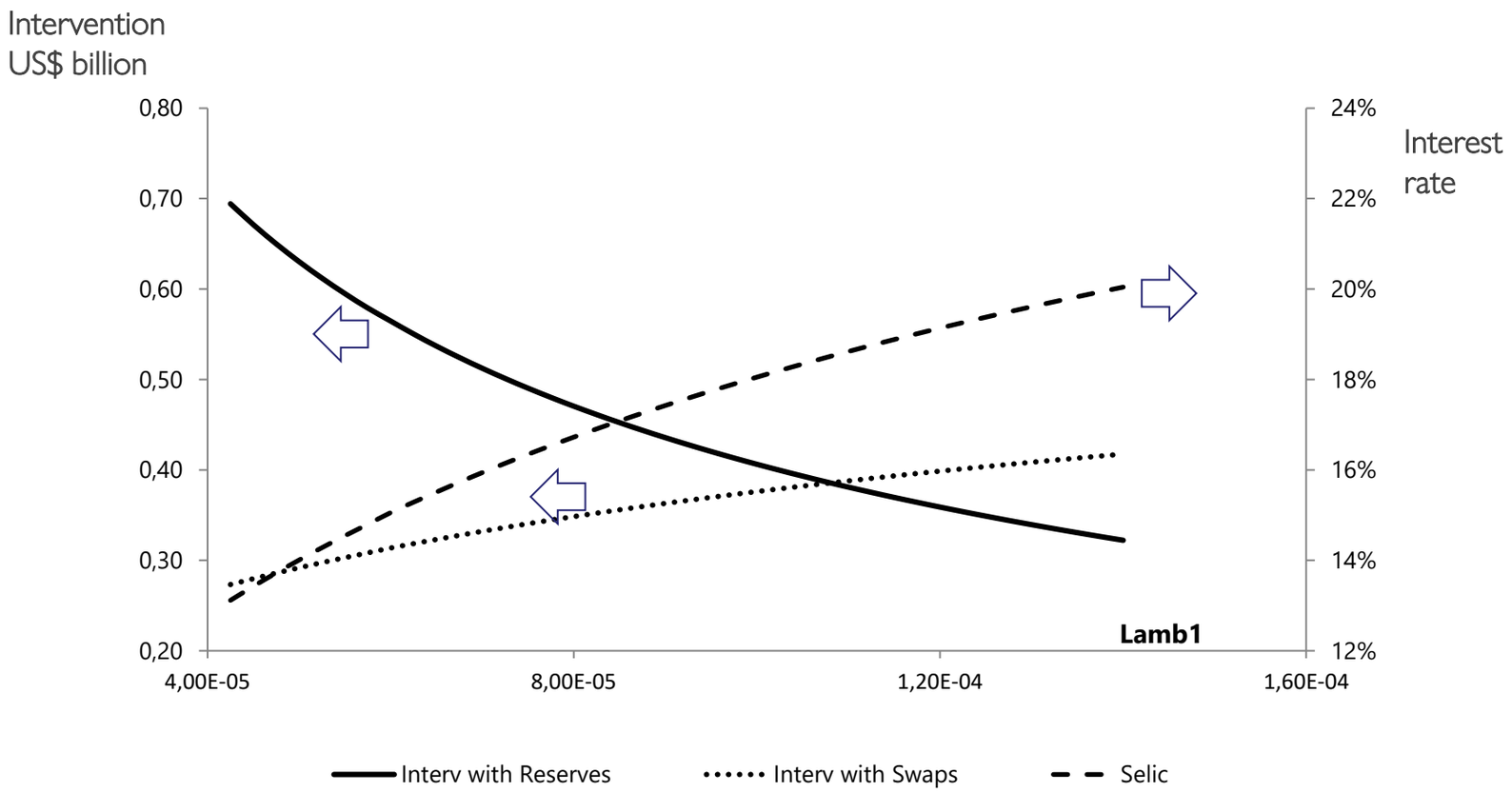
# Exchange rate stochastic process – model identification

| | Constant | Spot intervention | Swaps intervention | Interest rate |
|---|---|---|---|---|
| Coefficient | 0.002 | -0.0045 | -0.0025 | -0.011 |
| p-Value | 0.001 | 0.02 | 0.091 | 0.008 |

# Exchange rate stochastic process – calibration and sensitivity analysis

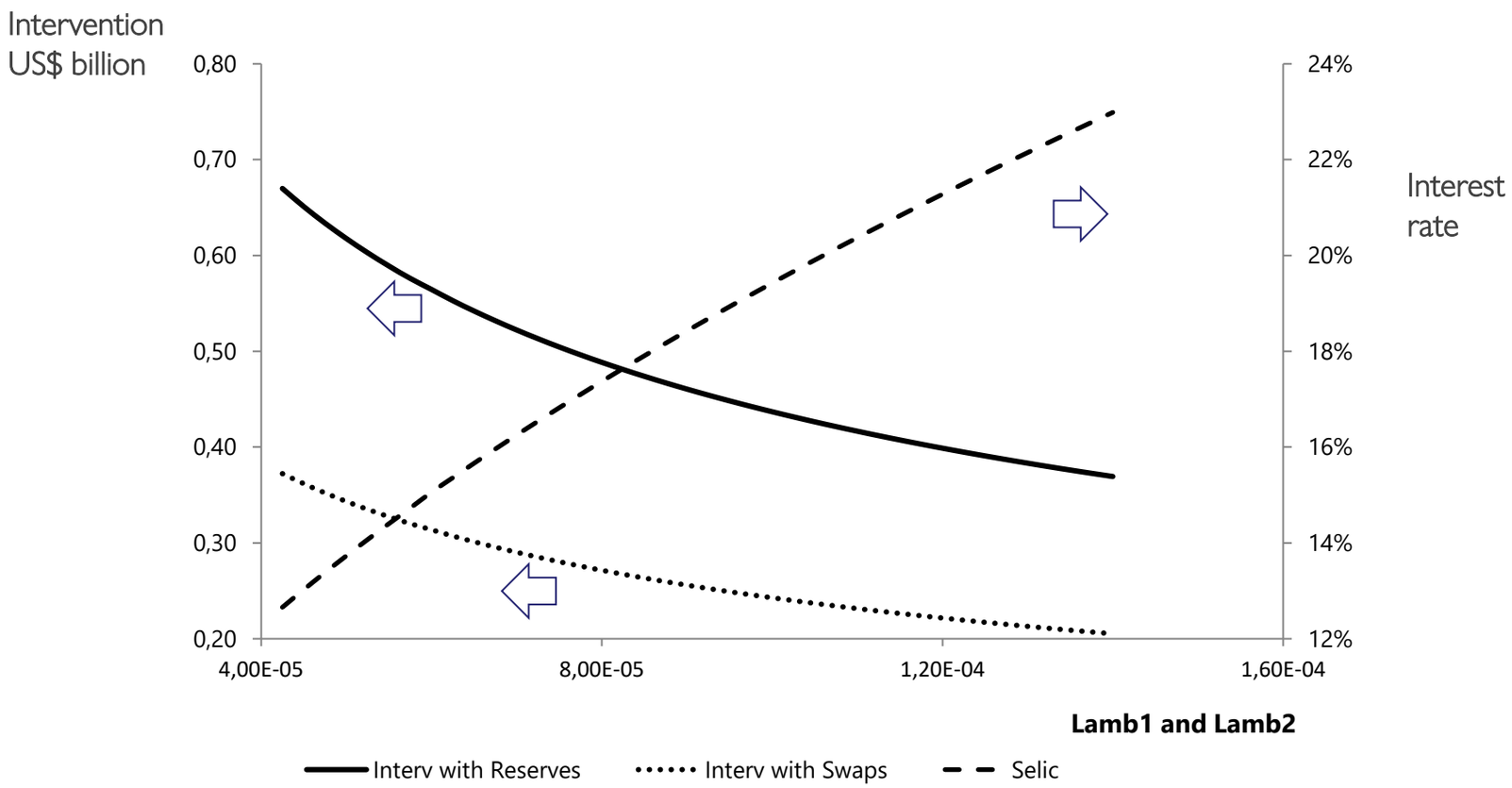
| *k* x 102 | *beta* x 104 | *lam1* x 105 | *lam2* x 105 | SELIC | *Qsi1* | *Qsi2* | PTAX | Case |
|---|---|---|---|---|---|---|---|---|
| 5 | 5.5 | 6 | 6 | 15.1 | 565 | 314 | 2.151 | Base |
| 1 | 5.5 | 6 | 6 | 15.1 | 564 | 313 | 2.248 | Flexibility with PTAX |
| 5 | 5.5 | 4 | 6 | 12.8 | 719 | 267 | 2.144 | High Lev Int Reserves |
| 5 | 5.5 | 8 | 6 | 16.7 | 470 | 348 | 2.156 | Low Lev Int Reserves |
| 5 | 5.5 | 6 | 4 | 14.2 | 533 | 445 | 2.148 | Low FX in the Debt |
| 5 | 5.5 | 6 | 8 | 15.5 | 582 | 242 | 2.152 | High FX in the Debt |
| 5 | 5.5 | 8 | 8 | 17.4 | 489 | 271 | 2.158 | Low Int Reserves High FX in the Debt |
| 5 | 5.5 | 4 | 4 | 12.3 | 691 | 384 | 2.142 | High Int Reserves Low FX in the Debt |

# Exchange rate stochastic process – scenarios




Lower flexibility with international reserves ⇨ More swaps and higher interest rate in interventions

# Exchange rate stochastic process – scenarios




Lower flexibility with international reserves AND swaps ⇨ Higher interest rate in interventions

# Exchange rate stochastic process

**Some conclusion of the model**

- Swaps help during crisis
- International reserve holdings allow lower interest rates
- BCB should reduce FX short exposure

# Exchange rate stochastic process – before 2008

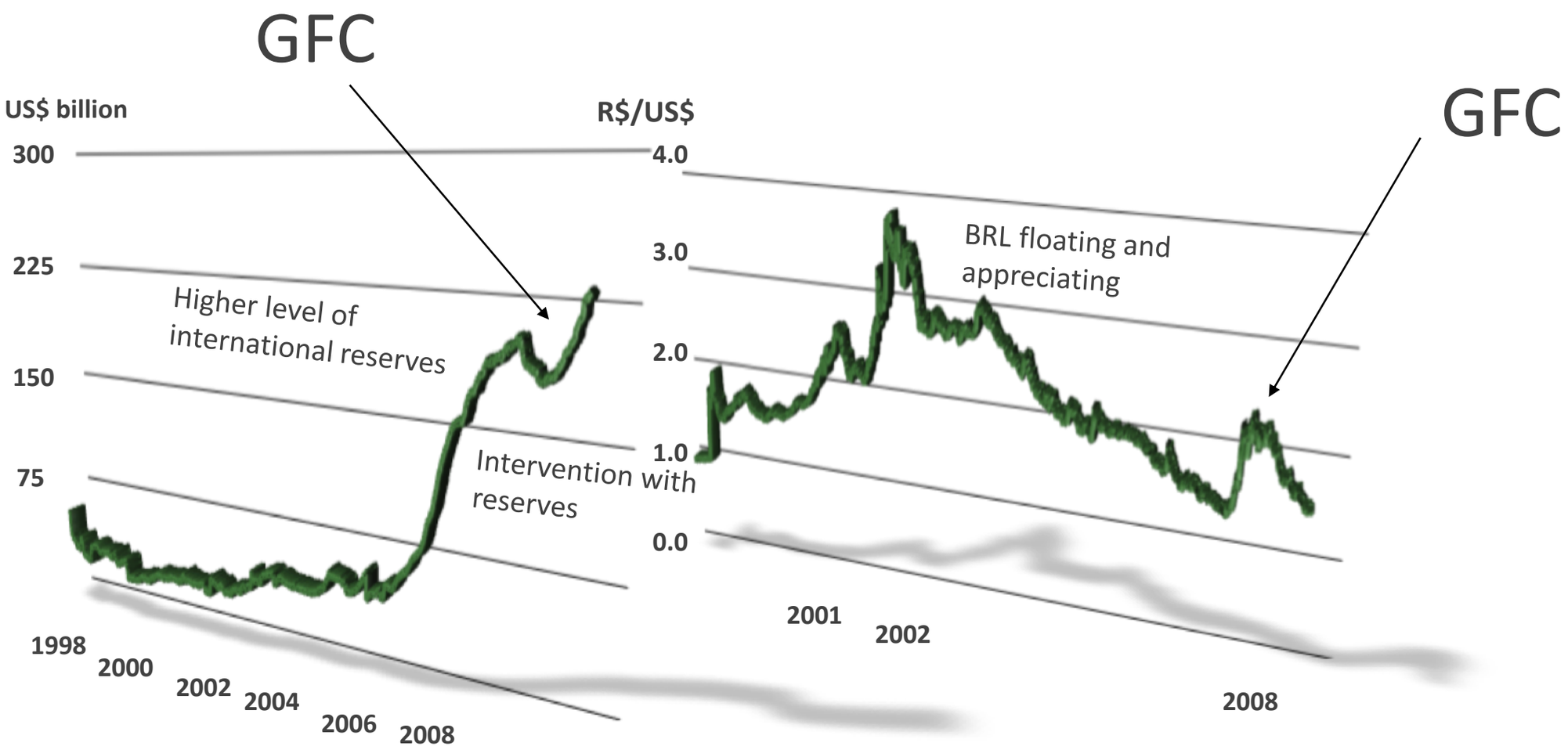
**Before 2008, the Brazilian central bank:**

– Increased international reserve holdings

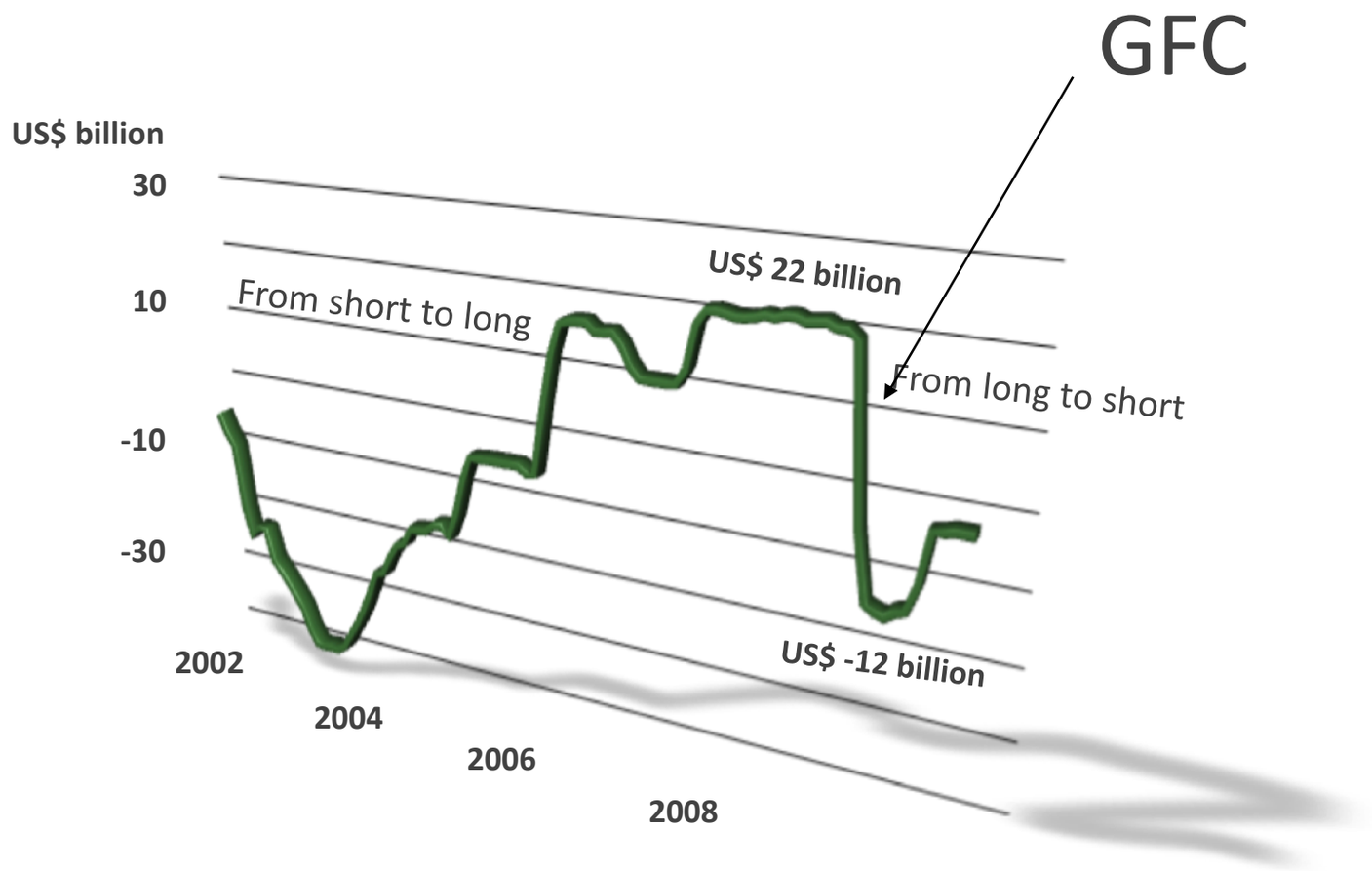
– Took long position in the swaps

The model suggests lower interest rate interventions in case of crisis.

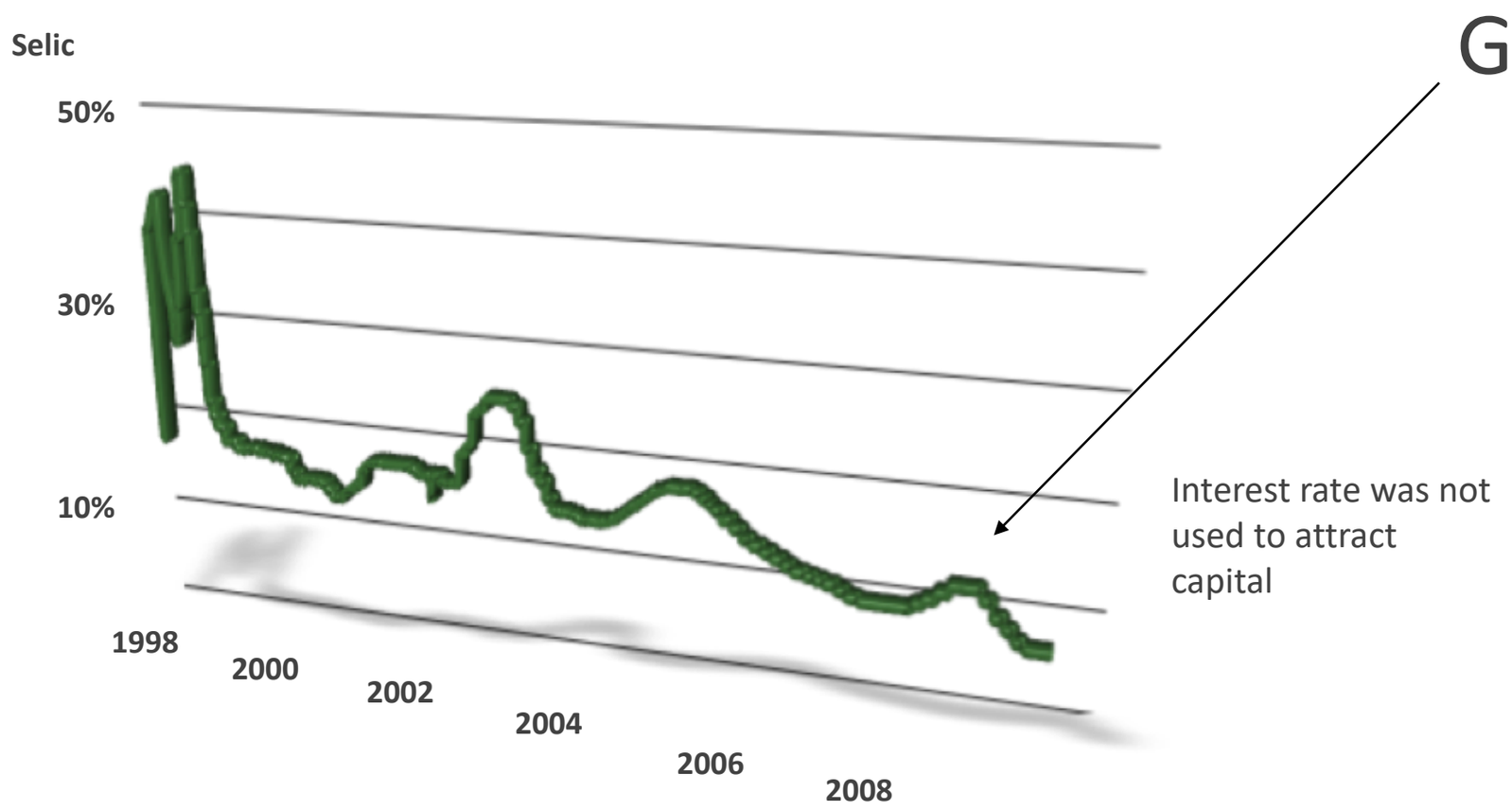
# International reserves and Exchange rate in the global financial crisis – GFC (2008-2010)



# Swaps and the global financial crisis (2008-2010)



# Exchange rate and the global financial crisis (2008-2010)



# Exchange rate and the global financial crisis (2008-2010)

| Instrument | 2001 | 2002 | 2008 |
|---|---|---|---|
| International reserves(US$ bi) | 8.2 | 7.4 | 14 |
| Swaps or Bonds (US$ bi) | 3.7 | 1.7 | 34 |
| Interest rate differential (%) | 16.4 | 18.1 | 13.5 |

# Exchange rate and the global financial crisis (2008-2010)

| | US$ (billion) |
|---|---|
| International reserves | 14 |
| Domestic swaps | 34 |
| Sell of US$ with repurchase agreement | 12 |
| US$ Loans with guarantees[1] | 13 |
| Total | 73 |

Interest rate (Not used to attract capital)

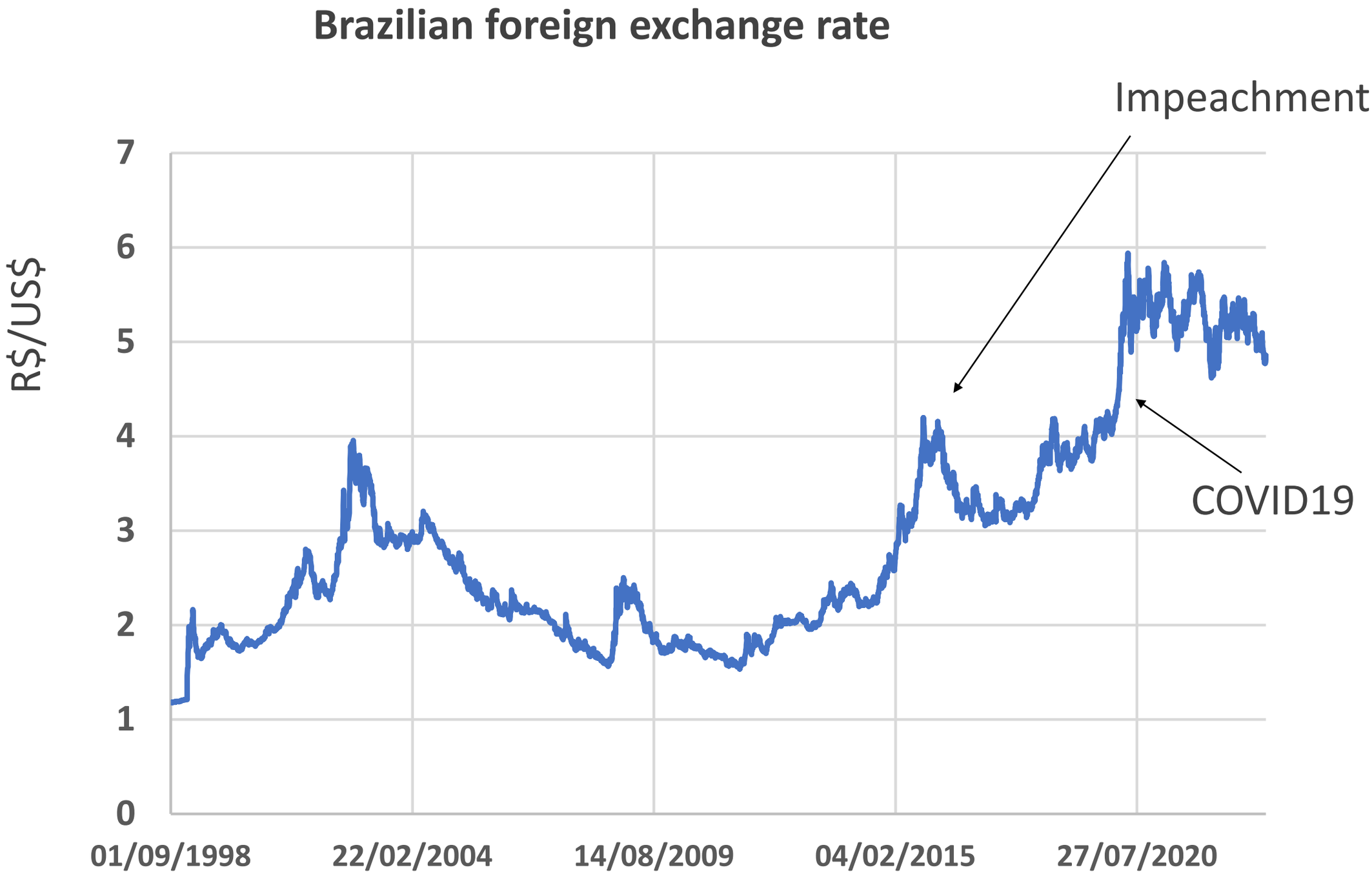
[1] US$ loans to financial institutions under agreement to lend to exporters. Collateral could be based on foreign exchange rate contracts and on exports contracts or global bonds.

# Exchange rate and the global financial crisis (2008-2010)

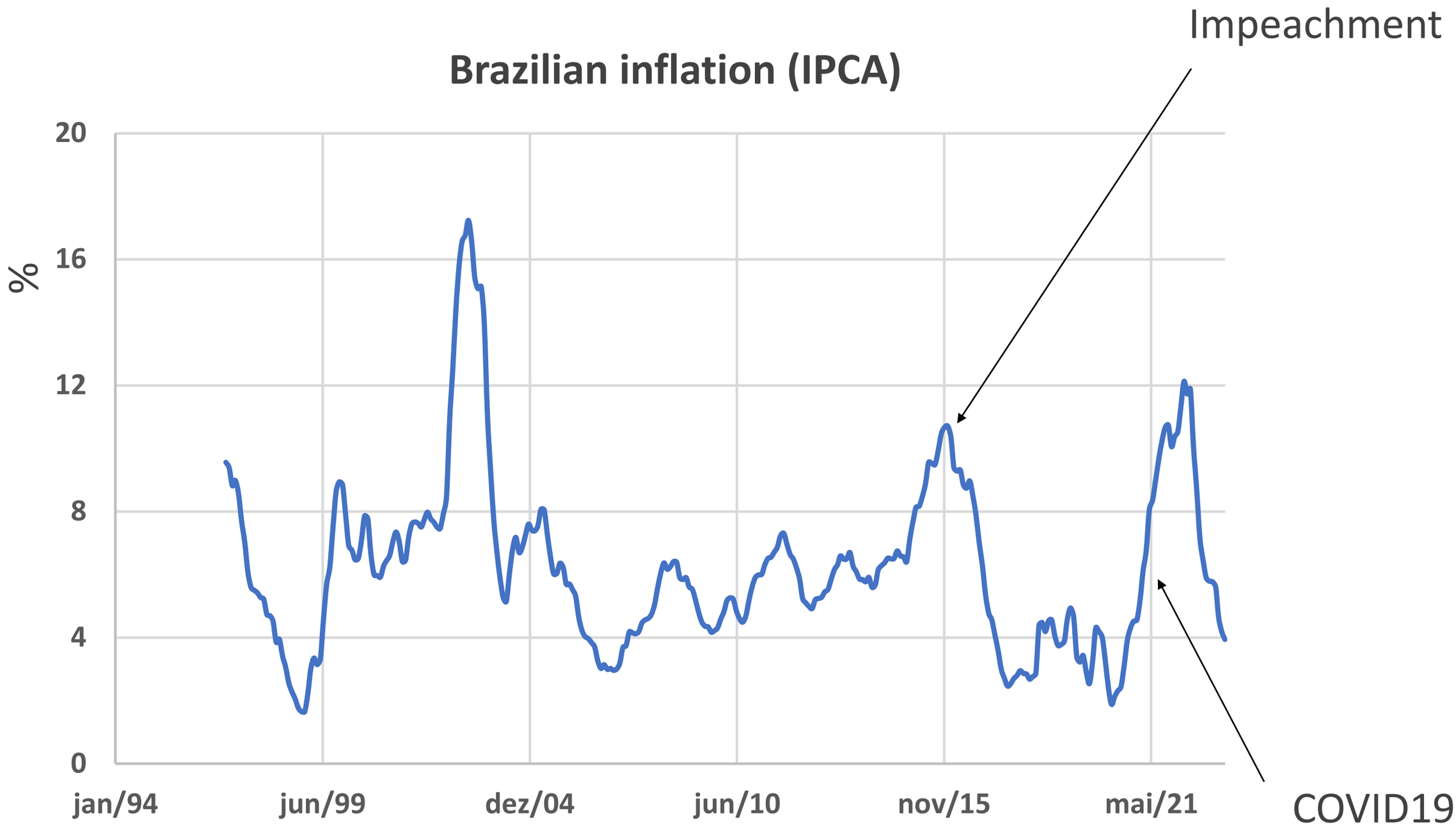
- Higher intervention flexibility allows the Central Bank of Brazil to intervene less with interest rate during a crisis

- The increase of international reserves level, the long position on swaps and the availability of others interventions tools helped the country to avoid higher interest rate in the 2008 crisis

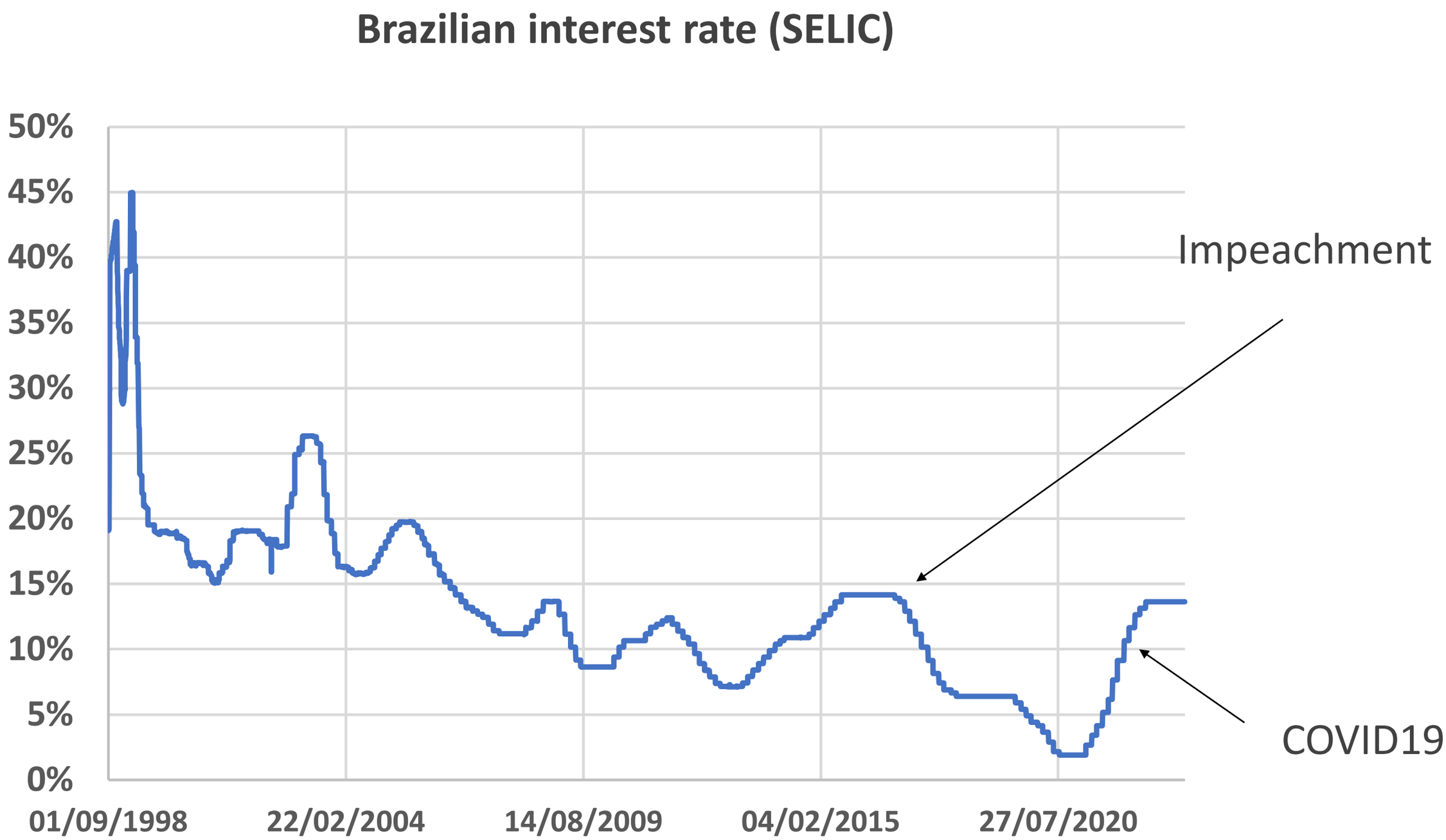
# Recent strategies in crises



# Recent strategies in crises



# Recent strategies in crises



# Recent strategies in crises



# Recent strategies in crises



# Questions

1. What are the main risks of the capital flows for emerging markets?
2. How monetary policy in the major economies affects emerging markets?
3. How international reserves may improve resilience of emerging markets?